24EECE2001 – Electronic Devices and Circuits

MOS FIELD EFFECT TRANSISTORS

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Syllabus

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MODULE 3 BIPOLAR JUNCTION TRANSISTORS Device Structure and Physical Operation, Current–Voltage Characteristics, BJT Circuits at DC, Applying the BJT in Amplifier Design, Small-Signal Operation and Models, Basic BJT Amplifier Configurations. Transistor Breakdown and temperature effects

MODULE 4 MOS FIELD EFFECT TRANSISTORS Device Structure and Physical Operation, Current—Voltage Characteristics, MOSFET Circuits at DC, Applying the MOSFET in Amplifier Design, Small-Signal Operation and Models, Basic MOSFET Amplifier Configurations, Biasing in MOS Amplifier Circuits, Discrete-Circuit MOS Amplifiers

MODULE 5 AMPLIFIER FREQUENCY RESPONSE Review of Transfer function, Frequency Response, Pole-Zero Concepts, Transfer function and AC Analysis of a First Order RC Circuit, Capacitive Effects in a PN Junction, Internal Capacitive Effects and the High-Frequency Model of the MOSFET, High-Frequency Response of the Common Source Amplifiers, High Frequency Response of Common Gate Amplifiers, High Frequency Response of Source Followers

Introduction to MOSFETs

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The basic principle involved in BJT/MOSFET is the voltage between two terminals can be to control the current flowing in the third terminal. Both FETs and Transistors can be used as signal amplifiers, electronic switches etc. but MOSFET has become by far the most widely used electronic device, especially in the design of integrated circuits (ICs), which are circuits fabricated on a single silicon chip.

Compared to BJTs, MOSFETs

- can be made quite small (i.e., requiring a small area on the silicon IC chip)
- manufacturing process is relatively simple
- operation requires comparatively little power.
- Furthermore, circuit designers have found ingenious ways to implement digital and analog functions utilizing MOSFETs almost exclusively (i.e., with very few or no resistors).

All of these properties have made it possible to pack large numbers of MOSFETs (>200 million!) on a single IC chip to implement very sophisticated, very-large-scale-integrated (VLSI) circuits such as those for memory and microprocessors.

The two major types of MOSFETs are Enhancement type and depletion type. Each of these types can be manufactured with *n*-channel or *p*-channel.

Device Structure (n-channel enhancement-type MOSFET)

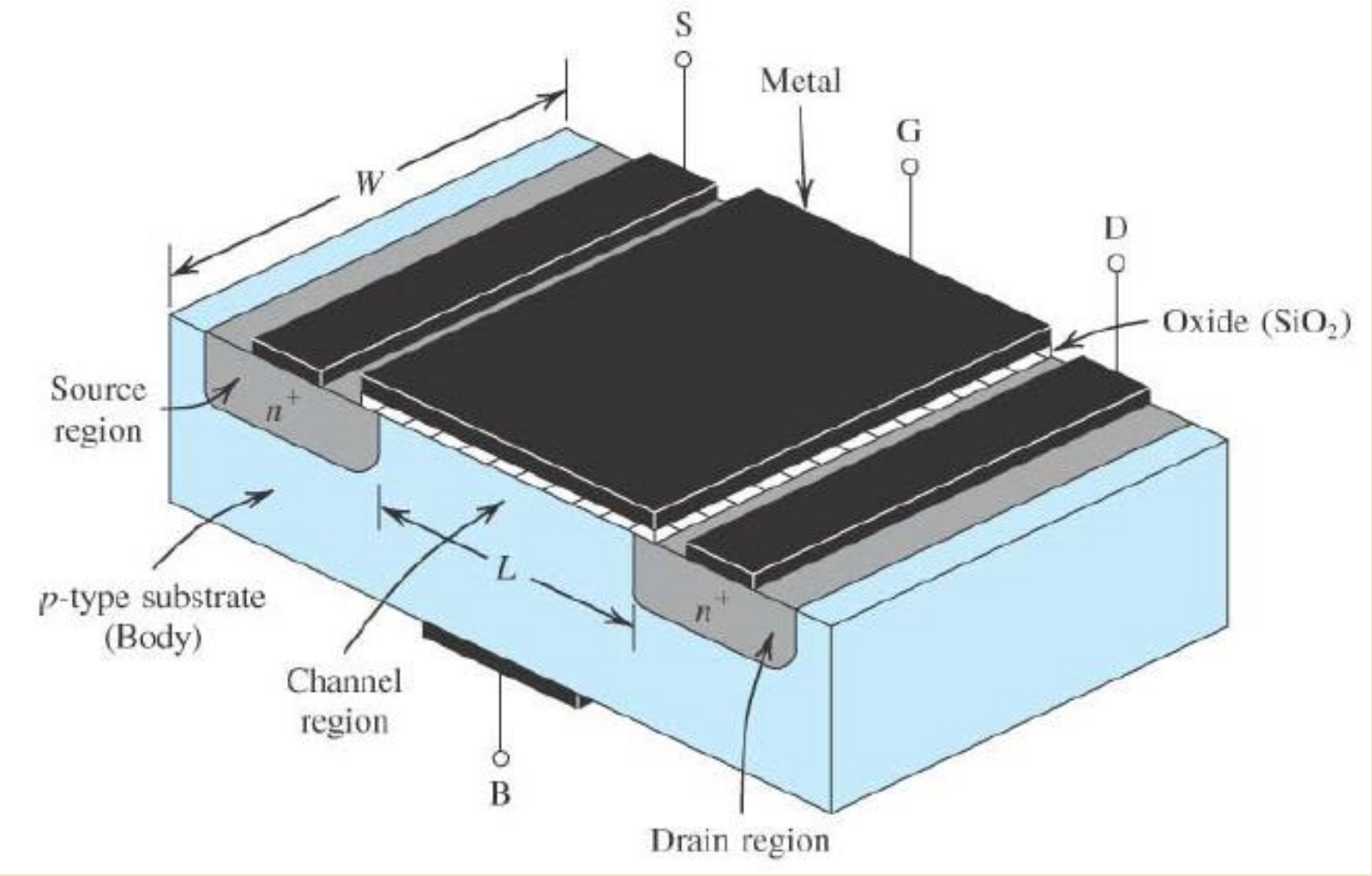


Figure 1 Physical structure of the enhancement-type NMOS transistor (perspective view)

The enhancement type MOSFET is most widely used FET. It finds extensive use in VLSI circuits. The physical structure of this type of MOSFET is shown below (Fig.1). The transistor is fabricated on a p-type substrate, which is a single-crystal silicon wafer that provides physical support for the device. Two heavily doped *n*-type regions (indicated with n^+ source and the

n^+ drain regions, are created (or implanted into the p-type substrate) in the substrate. A thin layer of silicon dioxide (S_iO_2) of thickness t_{ox} (typically 2-50 nm), which is an excellent electrical insulator, is grown on the surface of the substrate, covering the area between the source and drain regions. Metal is deposited on top of the oxide layer to form the gate electrode of the device. Metal contacts are also made to the source region, the drain region, and the substrate, also known as the body.

Thus four terminals are brought out: the gate terminal (G), the source terminal (S), the drain terminal (D), and the substrate or body terminal (B). Generally, body and source terminals are connected together forming a three terminal device. The region between two *n*-regions is called the channel. Typical values of channel length, *L* are 0.1 to 3 *μm* while channel width is typically 0.2 to 100 *μm*.

Physical structure of the enhancement-type NMOS transistor (cross-section)

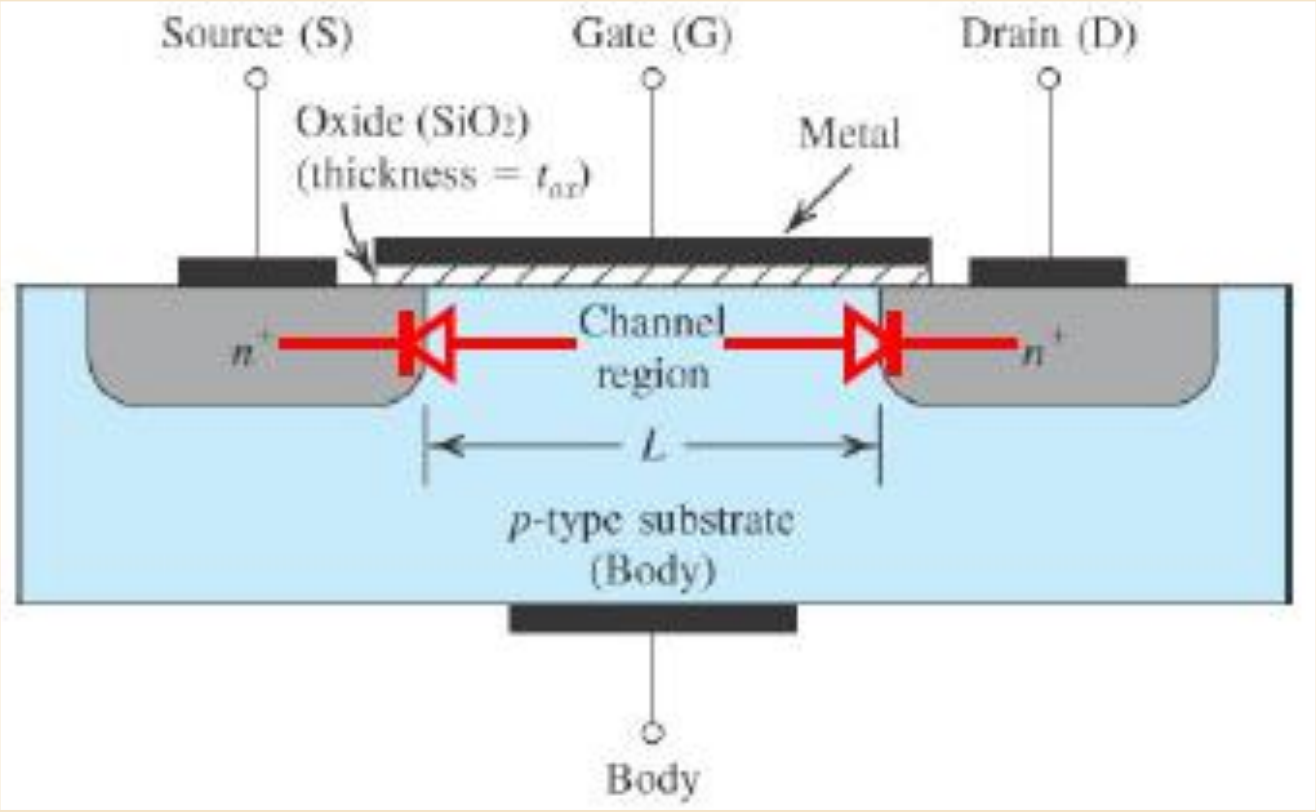
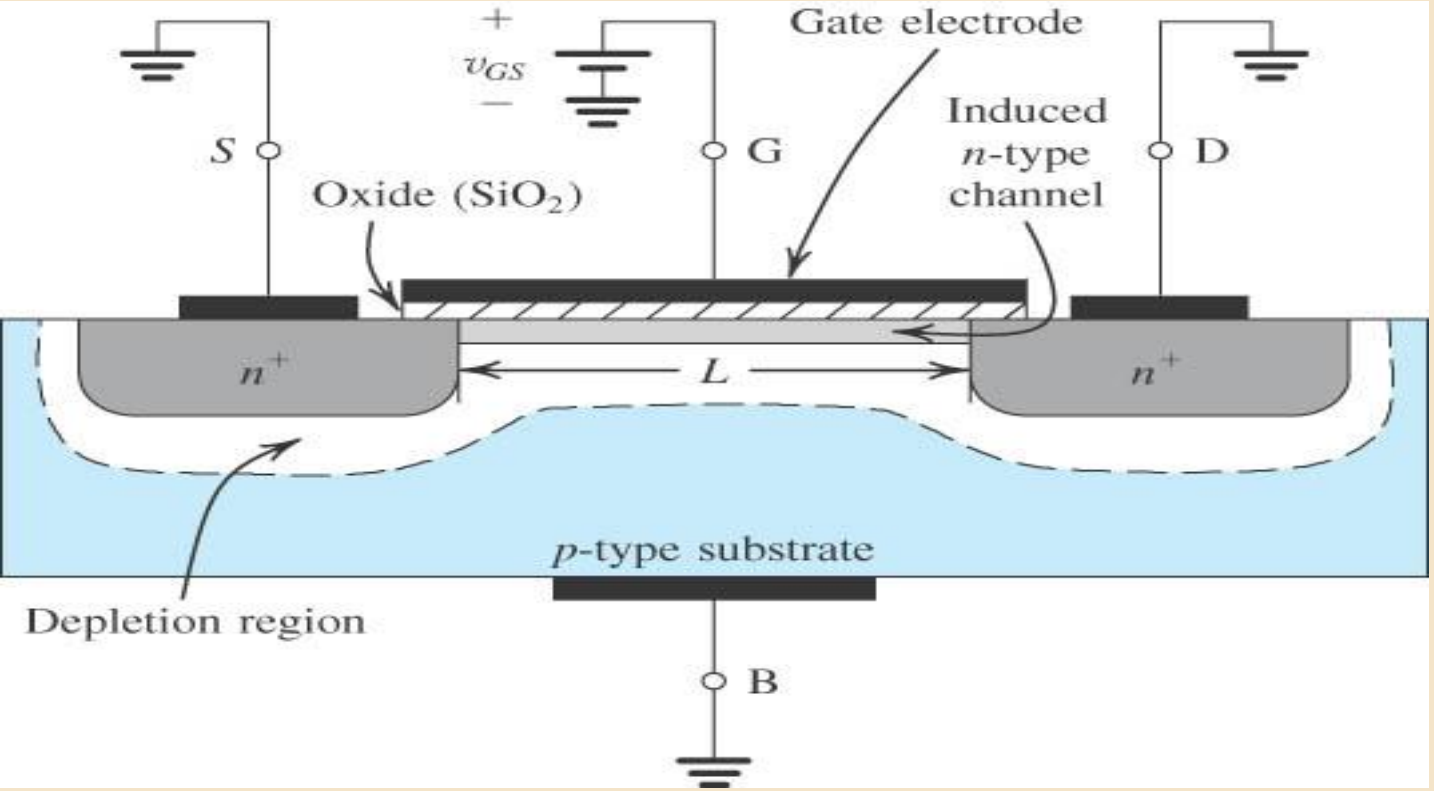


Figure 2 Physical structure of the enhancement-type NMOS transistor (cross-section)

Operation with no gate voltage:

With no bias voltage applied to the gate, two back-to-back diodes exist in series between drain and source. One diode is formed by the *pn* junction between the n^+ drain region and the p-type substrate, and the other diode is formed by the *pn* junction between the *p*-type substrate and the n^+ source region. These back-to-back diodes prevent current conduction from drain to source when a voltage $\boldsymbol{v_{DS}}$ is applied (current cannot flow into channel either from drain or from source) because the resistance between drain and source will be of the order of 10^{12} Ω).

Creating a Channel for Current Flow:



In order to obtain current flow, the MOSFETs needs to be biased and so a positive voltage is applied at the gate electrode and the other two terminals (source and drain) are grounded. Therefore the positive voltage, v_{GS} appears between gate and source.

Fig. 3 The enhancement-type NMOS transistor with a positive voltage applied to the gate.

An *n* channel is induced at the top of the substrate beneath the gate

The induced n region thus forms a channel for current flow from drain to source. Correspondingly, the MOSFET of Fig. 3 is called an **n-channel MOSFET** or, alternatively, an NMOS transistor. Note that an n-channel MOSFET is formed in a p-type substrate: The channel is created by inverting the substrate surface from p type to n type. Hence the induced channel is also called an **inversion layer.**

The value of v_{GS} at which a sufficient number of mobile electrons accumulated in the channel region to form a conducting channel is called the threshold voltage and is denoted V_t.

The value of V_t is typically lies in the range 0.5V to 1.0 V.

Applying a Small v_{DS}:

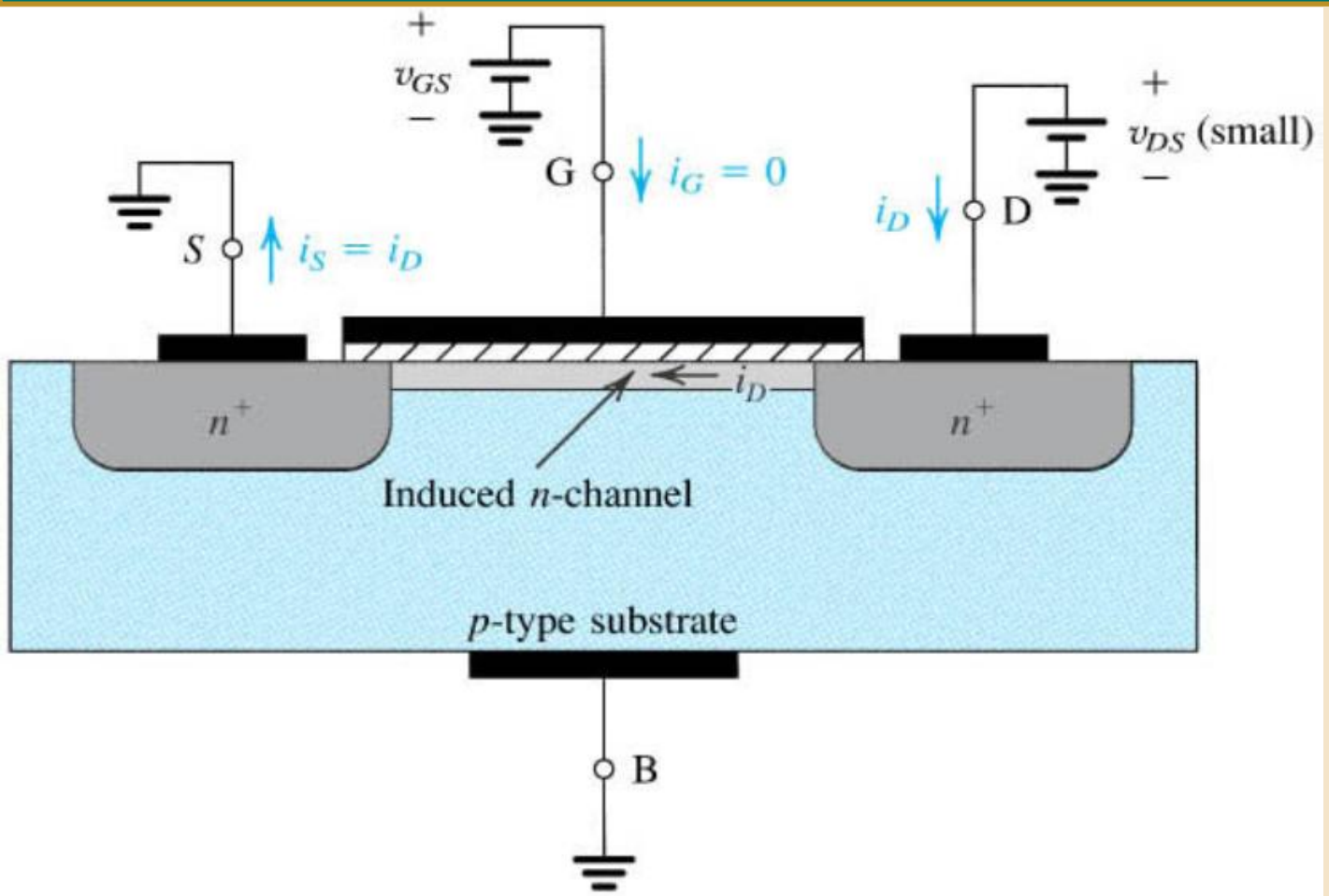


Figure 4 An NMOS transistor with $v_{GS} > V_t$ and with a small v_{DS} applied

All current entering the drain will exit the source. Therefore $i_S = i_D$.

Channel is induced when vGS - V_t > 0. If we increase v_{GS}, v_{GS} - V_t (excess gate voltage, also known as effective voltage or the overdrive voltage) increases, more electrons are attracted into the channel, which results in a higher channel conductivity (lower channel resistivity).

This process, of increasing the induced channel conductivity by increasing the excess gate voltage, is otherwise known as **channel enhancement** and hence the name **enhancement-type MOSFET**.

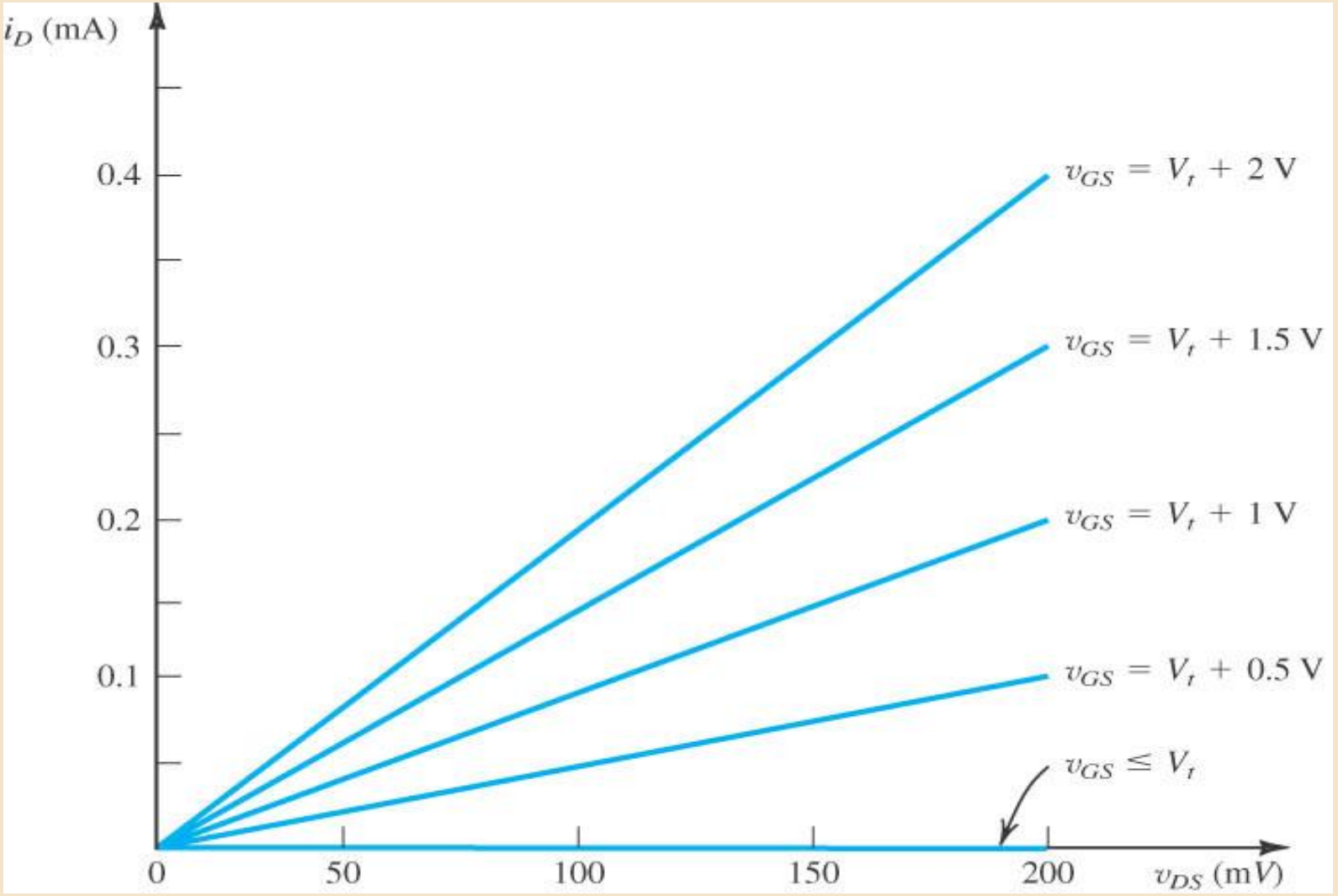
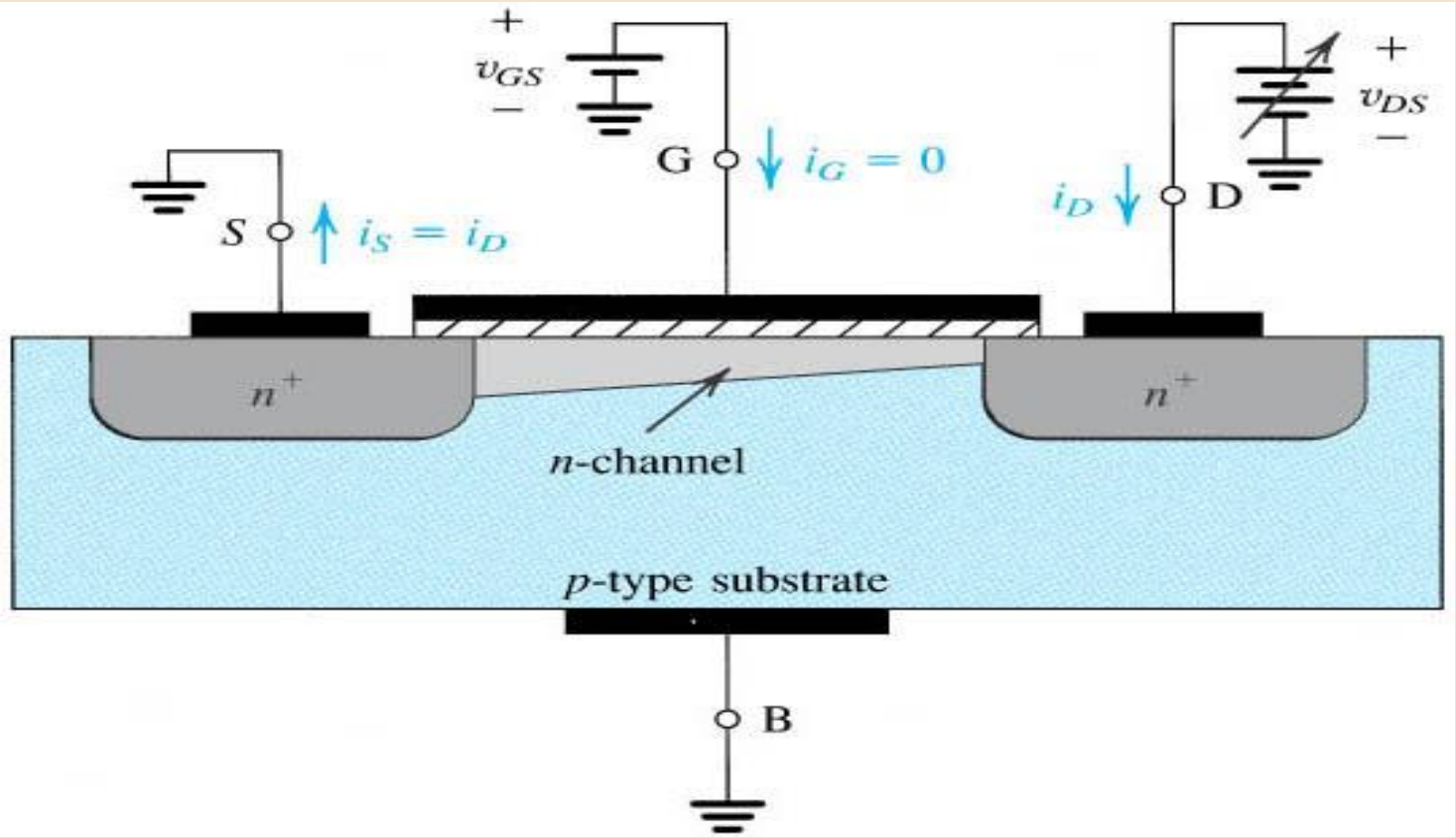


Figure 5 The i_D–v_{DS} characteristics of the MOSFET. when the voltage applied between drain and source, $v_{DS,}$ is kept small. The device operates as a linear resistor whose value is controlled by v_{GS}.

Operation as v_{DS} is increased

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Operation as v_{DS} is increased:



$$v_{GS} - v_{GD} - v_{DS} = 0$$

$$v_{DS} = v_{GS} - v_{GD}$$

If v_{DS} becomes sufficiently large that $v_{GD} < V_t$, the voltage at drain end becomes small than at the source end. As the channel depth depends on this voltage (specifically on the amount by which this voltage exceeds V_t), the channel is no longer of uniform depth; rather the channel will take the tapered shape as in Fig. 6, being deepest at the source end and shallowest at the drain end.

Fig. 6 Operation of the enhancement NMOS transistor as v_{DS} is increased

The pinch-off voltage in a MOSFET is the drain-source voltage at which the channel near the drain narrows and the device enters saturation, limiting the drain current.

$$v_{DSat} = v_{GS} - V_t$$

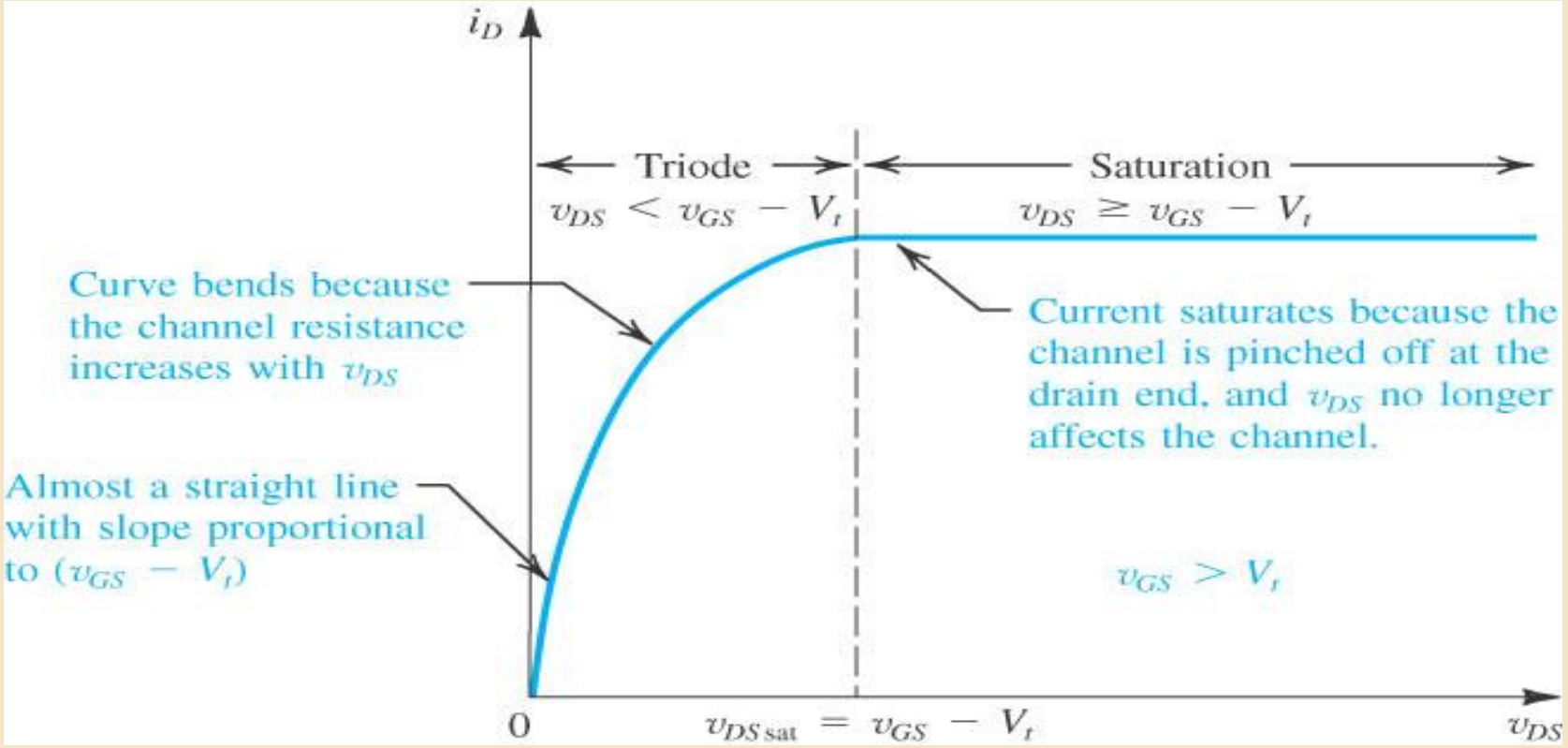


Fig. 7 The drain current i_D versus the drain-to-source voltage v_{DS} for an enhancement-type NMOS transistor operated with $v_{GS} > V_t$

When $v_{GD} = V_t$, or $v_{DS} = v_{GS} - V_t$, the channel depth at the drain end decreases to almost zero, and the channel is said to be pinched off. Increasing v_{DS} beyond this value has little effect (theoretically, no effect) on the channel shape, and the current through the channel remains constant at the value reached for $v_{DS} = v_{GS} - V_t$. The drain current thus saturates at this value, and the MOSFET is said to have entered the saturation region of operation. The voltage v_{DS} at which saturation occurs denoted v_{DSat},

$$v_{DSat} = v_{GS} - V_t$$

The region of the i_D-v_{DS} characteristic obtained for $v_{DS} < v_{DSat}$ is called the **triode region** and the device operates in the **saturation region** if $v_{DS} > v_{DSat}$.

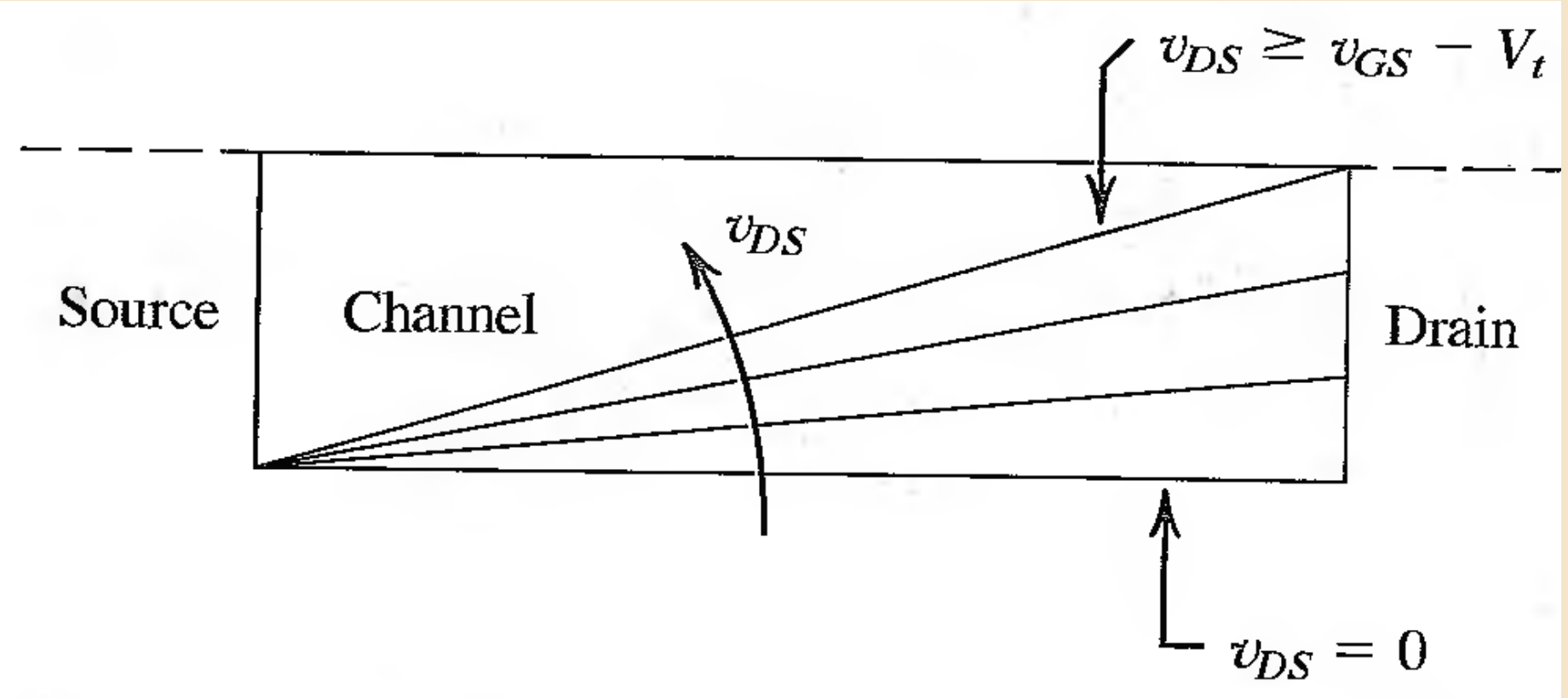


Fig. 8 Channel variation when v_{DS} is increased

Fig. 8 shows the sketches of channel when v_{DS} is increased while v_{GS} is kept constant. Increasing v_{DS} causes the channel to acquire a tapered shape. Eventually, as v_{DS} reaches v_{GS} - V_t, the channel is pinched off at the drain end:

Derivation of the i_D-v_{DS} Relationship

In MOSFET, the gate and the channel region form a parallel plate capacitor and the oxide layer serves as a dielectric. If the capacitance per unit gate area is denoted as C_{ox} and the thickness of the oxide layer is t_{ox}, then

$$C_{ox} = \epsilon_{ox}/t_{ox}$$

Where ϵ_{ox} is the Permittivity of the silicon oxide, $\epsilon_{ox} = 3.9\epsilon_o = 3.9 \text{ X } 8.854 \text{ X } 10^{-12} = 3.45 \text{ X } 10^{-11}$ F/m

The oxide thickness t_{ox} is determined by the process technology used to fabricate the MOSFET. Usually, t_{ox} = 10 nm, $C_{ox} = 3.45 \text{ X } 10^{-3}$ F/m^2 or 3.45 **fF/µm²** (fF represents femtofarad which is equivalent to 10^{-15}F).

Total capacitance between gate and channel is $C = C_{ox}WL$ --- (1)

where W is the width of the channel and L is the length of the channel.

The magnitude of the electron charge in the channel is $|Q| = C_{ox}WL(V_{GS} - V_t)$

Charge per unit channel length can be written as $\dfrac{|Q|}{unit\ channel\ length} = C_{ox}W(V_{GS} - V_t)$

The voltage v_{DS} establishes an electric field E across the length of the channel,

$$|E| = \frac{v_{DS}}{L} \qquad (4)$$

This electric field in turn causes the channel electrons to drift toward the drain with a velocity given by

$$Electron\ drift\ velocity = \mu_n |E| = \mu_n \frac{v_{DS}}{L} \qquad (5)$$

where μ_n is the mobility of electrons in the channel (called surface mobility). It is a physical parameter whose value depends on the fabrication process technology.

The value of i_D = charge per unit channel length x electron drift velocity

$$\text{Therefore} \quad i_D = [(\mu_n C_{ox})\frac{W}{L}(V_{GS} - V_t)]v_{DS} \qquad (6)$$

Thus, for small values of v_{DS}, the channel behaves as a linear resistance whose value is controlled by excess gate voltage, (v_{GS} - V_t), which in turn is determined by v_{GS}.

In the above equation, the first term ($\mu_n C_{ox}$), is determined by the process technology used to fabricate the MOSFET, which is termed the **process transconductance parameter** denoted by k_n where the subscript *n* denotes *n*- channel. (ref: i_D = v_{DS} x conductance).

The second term in the above equation, *W*/*L*, is called the aspect ratio of the MOSFET which is determined by the device engineer. The conductance is directly proportional to the channel width *W* and inversely proportional to the channel length *L*. This term is dimensionless.

The product of the **process transconductance parameter** k_n' **and** the aspect ratio of the MOSFET (*W*/*L*) is the MOSFET **transconductance parameter** k_n.

$$\therefore k_n = k_n'(W/L) \quad \text{or} \quad \therefore k_n = (\mu_n C_{ox})(W/L) \tag{7}$$

From equation (6), we can conclude that with v_{DS} kept small, the MOSFET behaves as a linear resistance r_{DS} whose value is controlled by the gate voltage, v_{GS}.

$$r_{DS} = \frac{1}{(\mu_n C_{ox})\frac{W}{L}(V_{GS} - V_t)} \tag{8}$$

Now consider when v_{DS} is increased. As v_{DS} is increased, the channel becomes more tapered and the resistance increases correspondingly. Thus, the i_D-v_{DS} curve does not continue a straight line but bends as shown in the characteristics. The charge in the tapered channel is proportional to the channel cross-sectional area. This cross sectional area is proportional to the voltage. So the average voltage (v_{GS} - V_t – (½)v_{DS})

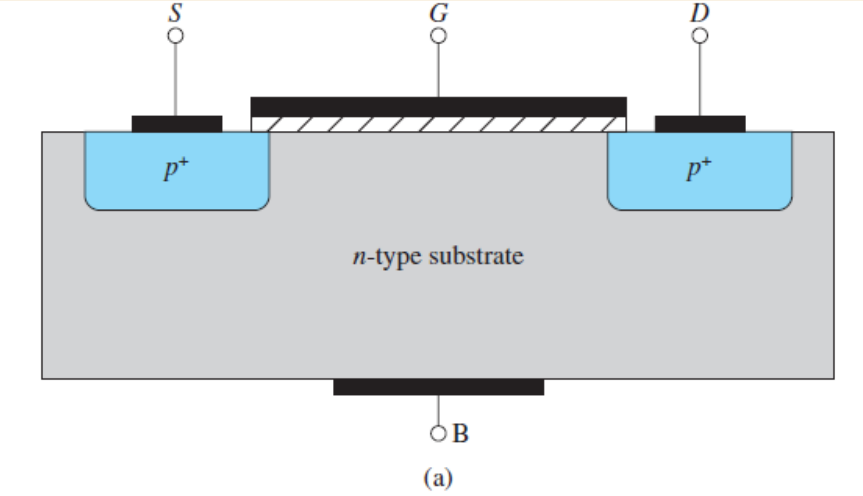
$$\text{Now } i_D = [(\mu_n C_{ox})\frac{W}{L}\left(V_{GS} - V_t - \frac{1}{2}v_{DS}\right)]v_{DS} = k_n^{'}(W/L)(V_{GS} - V_t)v_{DS} - \frac{1}{2}v_{DS}^2 \qquad (9)$$

The equation gives the expression for drain current in triode region.

When v_{DS} is further increased, $v_{DSat} = v_{GS} - V_t$.

$$i_D = \frac{1}{2}k_n^{'}(W/L)(V_{GS} - V_t)^2 \qquad (10)$$

The above equation gives the expression for drain current in saturation region.



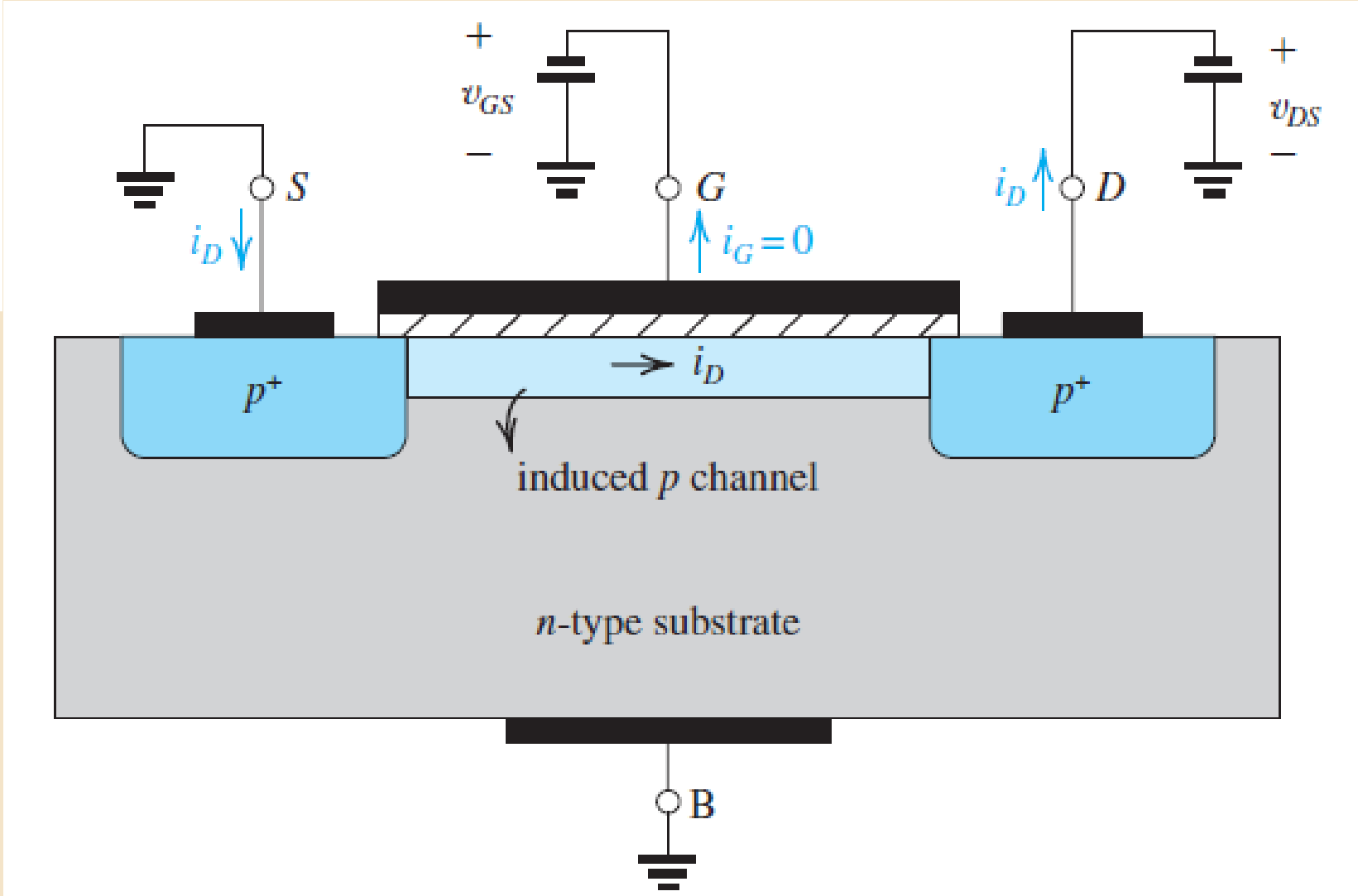
$$v_{GS} \leq V_{tp}$$

$$|v_{GS}| \geq |V_{tp}|$$

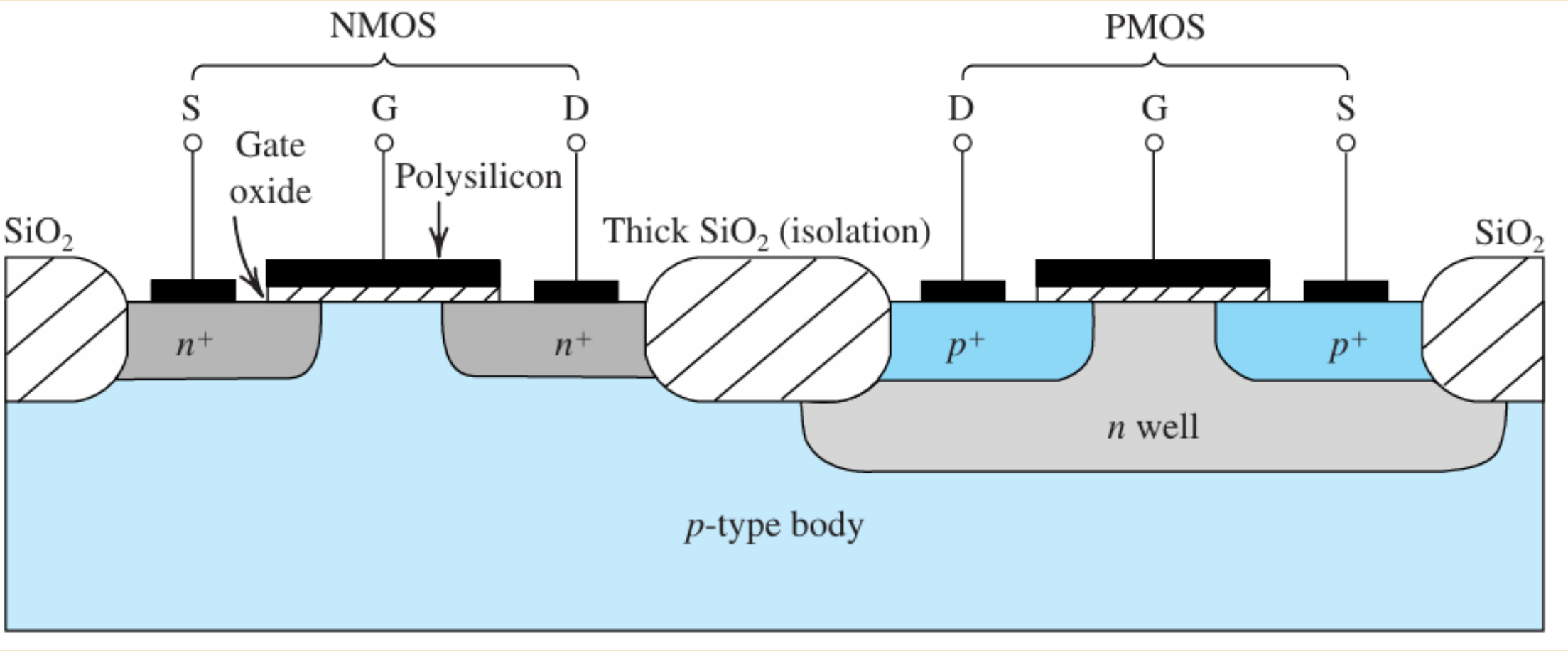
The process transconductance parameter for the PMOS device is

$$k_p' = (\mu_p C_{ox})$$

$$\therefore k_p = k_p' (W/L)$$



Complementary MOS or CMOS



NMOS transistor is implemented directly in the p-type substrate, the PMOS transistor is fabricated in a specially created n region, known as an **n well**. The two devices are isolated from each other by a thick region of oxide that functions as an insulator. Not shown on the diagram are the connections made to the p-type body and to the n well. The latter connection serves as the body terminal for the PMOS transistor.

Operating the MOS Transistor in the Subthreshold Region:

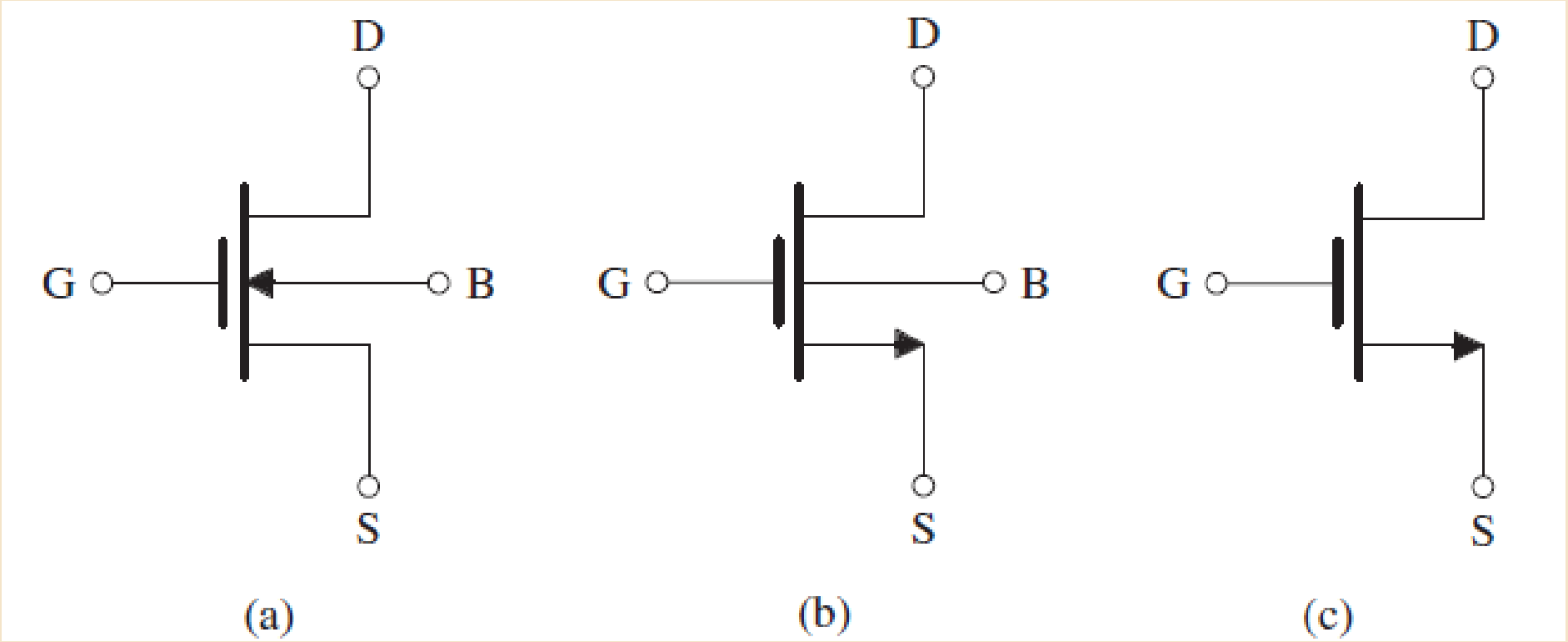
In n-channel MOSFET, when $v_{GS} < V_t$, no current flows and the device is cut off. But this is not entirely true, a small drain current flows for values of v_{GS} smaller than but close to V_t. In this subthreshold region of operation the drain current is exponentially related to v_{GS}.

Although in most applications the MOS transistor is operated with $v_{GS} > V_t$, there are special, but a growing number of, applications that make use of subthreshold operation.

Current-Voltage Characteristics

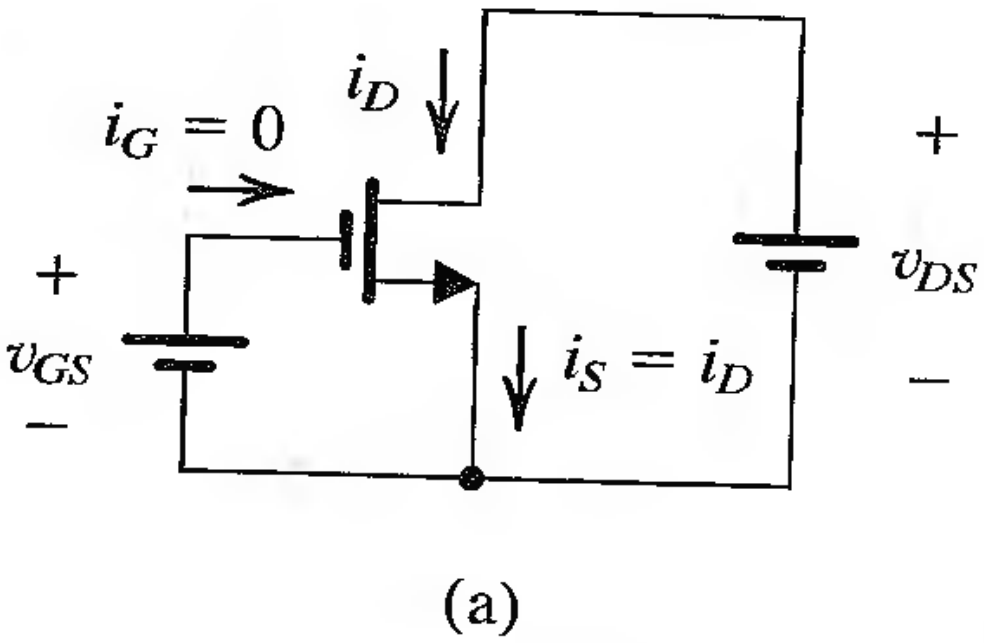
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Circuit symbol (enhancement NMOS)

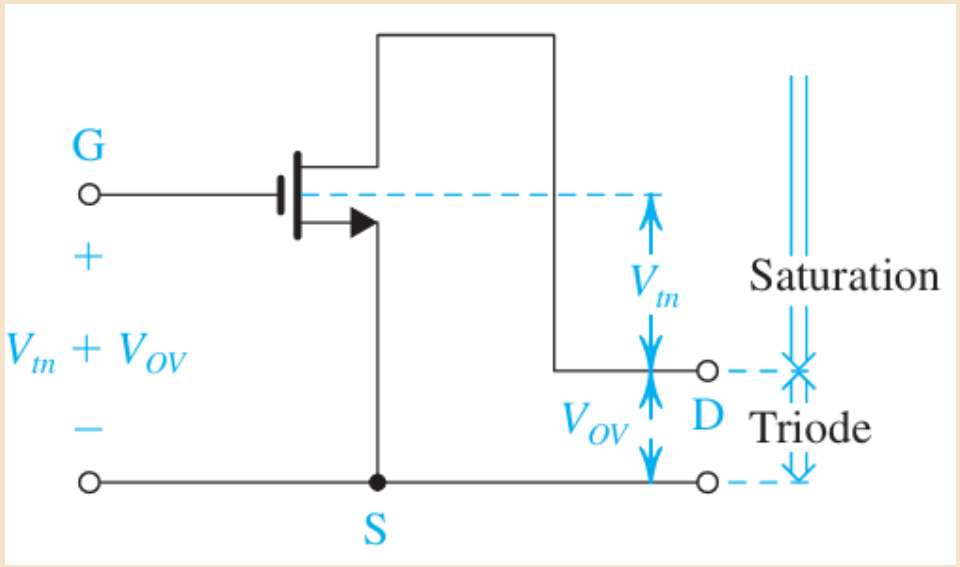


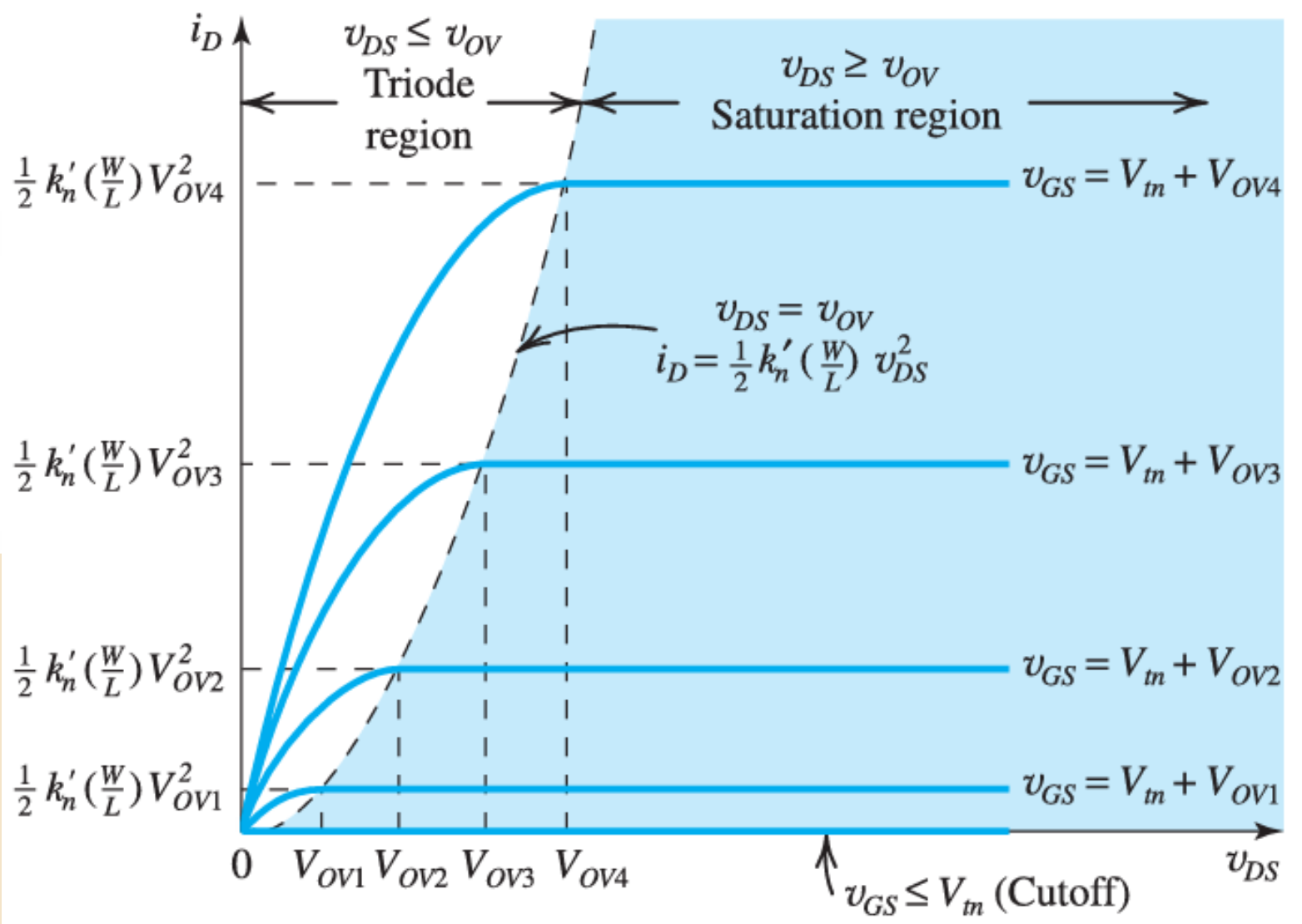
(a) Circuit symbol for the *n*-channel enhancement-type MOSFET
(b) Modified circuit symbol with an arrowhead on the source terminal to distinguish it from the drain
(c) Simplified circuit symbol to be used when the source is connected to the body

The i_D-v_{DS} Characteristics for an enhancement-type NMOS transistor

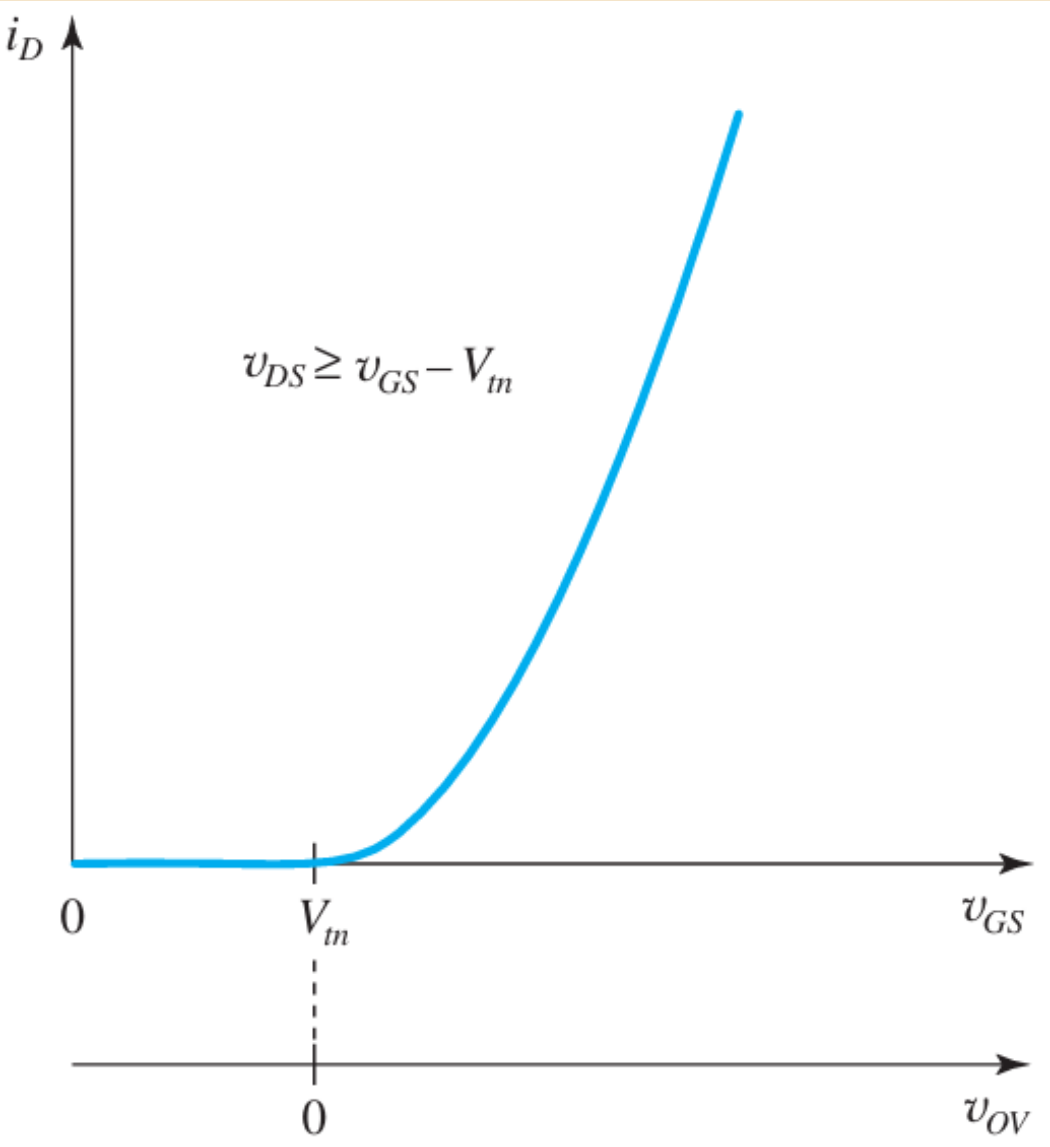


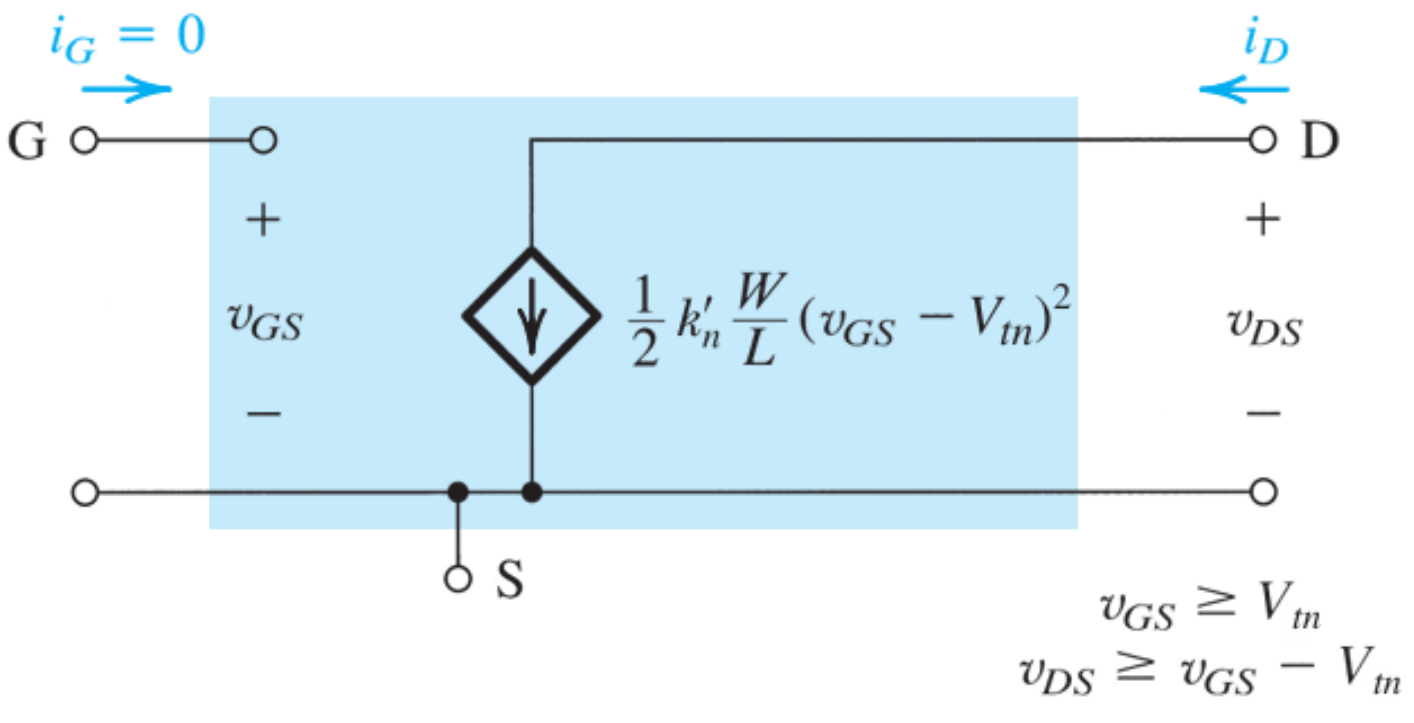
(a)





The I_D–v_{GS} Characteristic

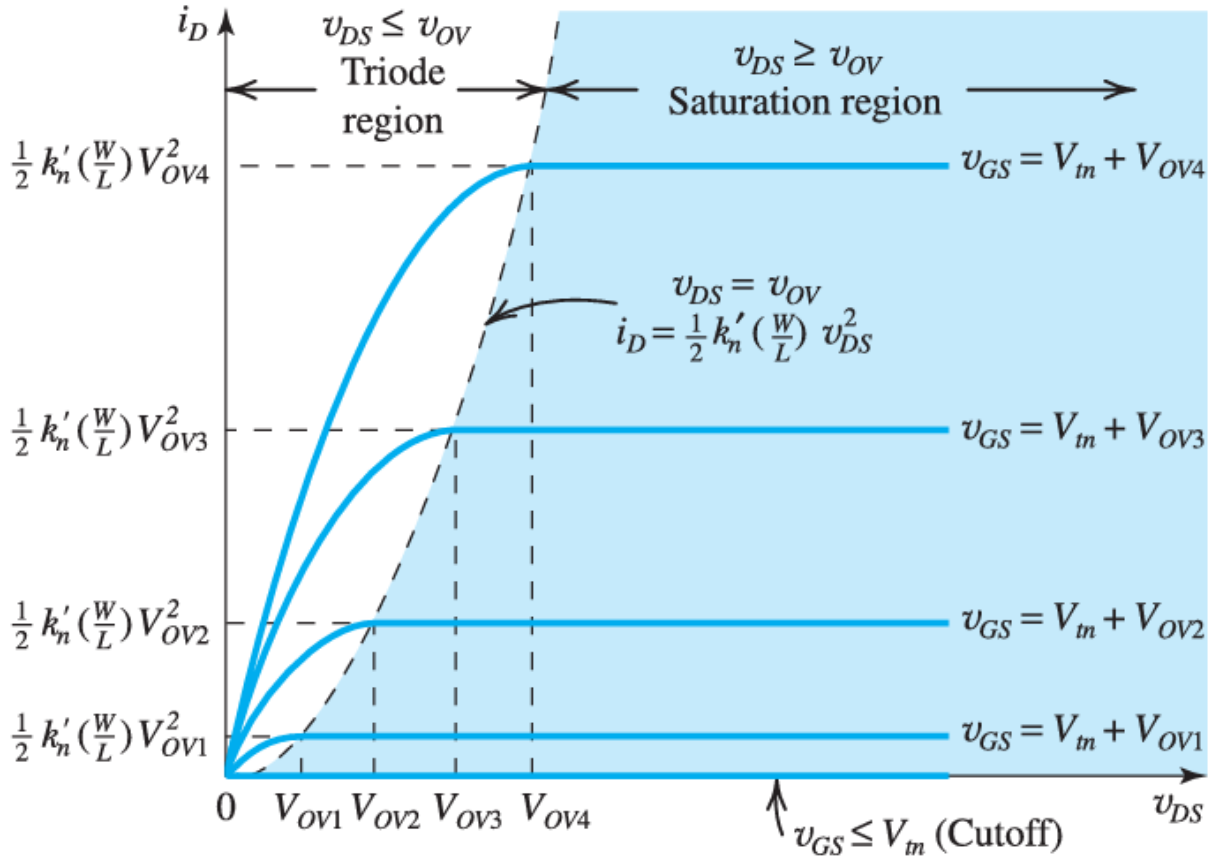




Large-signal, equivalent-circuit model of an n-channel MOSFET operating in the saturation region

i_D–v_{GS} characteristic of an NMOS transistor operating in saturation.

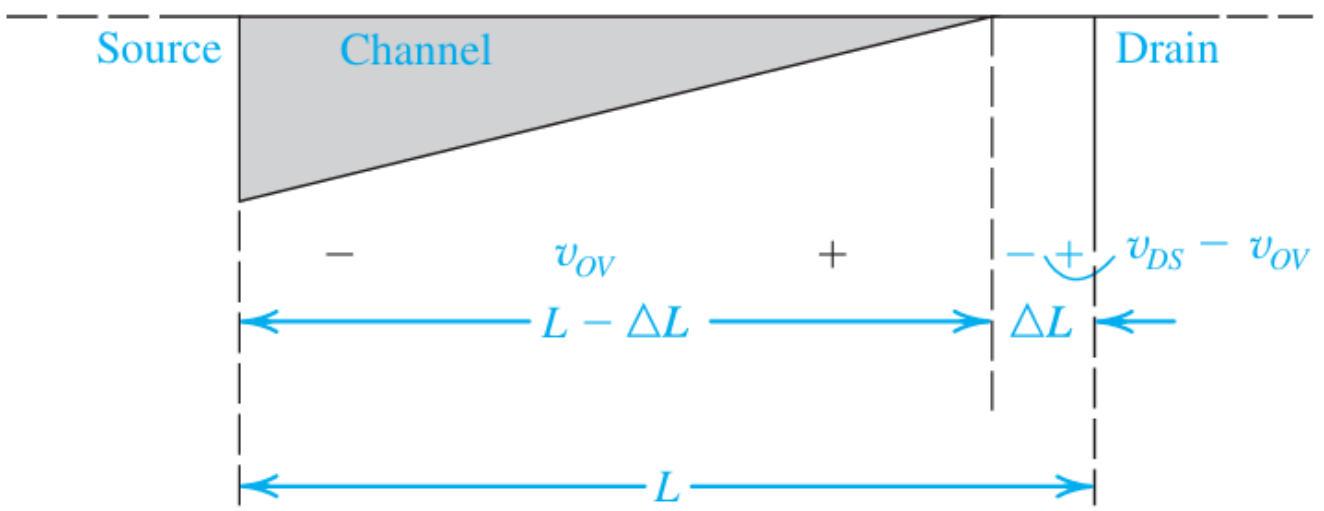
Channel-length Modulation and its effect on Output Resistance



$$i_D = \frac{1}{2}k_n'\frac{W}{L}(v_{GS} - V_t)^2$$

The above equation and the graph indicate that in saturation, i_D is independent of v_{DS}. Thus, a change Δv_{DS} in the drain-to-source voltage causes a zero change in i_D, which implies that the incremental resistance looking into the drain of a saturated MOSFET is infinite. i.e, once the channel is pinched off at the drain end, further increases in v_{DS} have no effect on the channel's shape.

But, in practice, as v_{DS} is increased, the channel pinch-off point is moved slightly away from the drain, toward the source.



The voltage across the channel remains constant at $v_{GS} - V_t = v_{DSat}$, and the additional voltage applied to the drain appears as a voltage drop across the narrow dv_{DS}epletion region between the end of the channel and the drain region.

This voltage accelerates the electrons that reach the drain end of the channel and sweeps them across the depletion region into the drain. Note, however, that (with depletion-layer widening) the channel length is in effect reduced, from L to $L-\Delta L$, a phenomenon known as **channel-length modulation**. Now, since i_D is inversely proportional to the channel length, i_D increases with v_{DS}.

To find the effect of i_D on v_{DS} in saturation, replace L with $L-\Delta L$ in equation

In this equation, assumed that $\Delta L / L \ll 1$. Assuming ΔL is proportional to v_{DS}, $\Delta L = \lambda' v_{DS}$

where λ' is a process-technology parameter with the dimensions of μm/V. Therefore

$$i_D = \frac{1}{2} k_n' \frac{W}{L}\left(1 + \frac{\lambda'}{L} v_{DS}\right)(v_{GS} - V_t)^2$$

$$i_D = \frac{1}{2} k_n' \frac{W}{L-\Delta L}(v_{GS} - V_t)^2$$

$$= \frac{1}{2} k_n' \frac{W}{L} \frac{1}{1-(\Delta L/L)}(v_{GS} - V_t)^2$$

$$\cong \frac{1}{2} k_n' \frac{W}{L}\left(1 + \frac{\Delta L}{L}\right)(v_{GS} - V_t)^2$$

Definition: Channel-length modulation is the reduction in effective MOSFET channel length with increasing v_{DS}, causing the drain current to increase slightly in the saturation region.

$$i_D = \frac{1}{2} k_n' \frac{W}{L}\left(1 + \frac{\lambda'}{L} v_{DS}\right)(v_{GS} - V_t)^2$$

Usually, λ'/L is denoted λ,

$$\lambda = \frac{\lambda'}{L}$$

Here, λ is a process process-technology parameter (V^{-1}) and is inversely proportional to the length selected for the channel. Now, i_D becomes

$$i_D = \frac{1}{2} k_n' \frac{W}{L}(v_{GS} - V_t)^2(1 + \lambda v_{DS})$$

in the above equation, i_D depends linearly on v_{DS} by a factor $(1+\lambda v_{DS})$

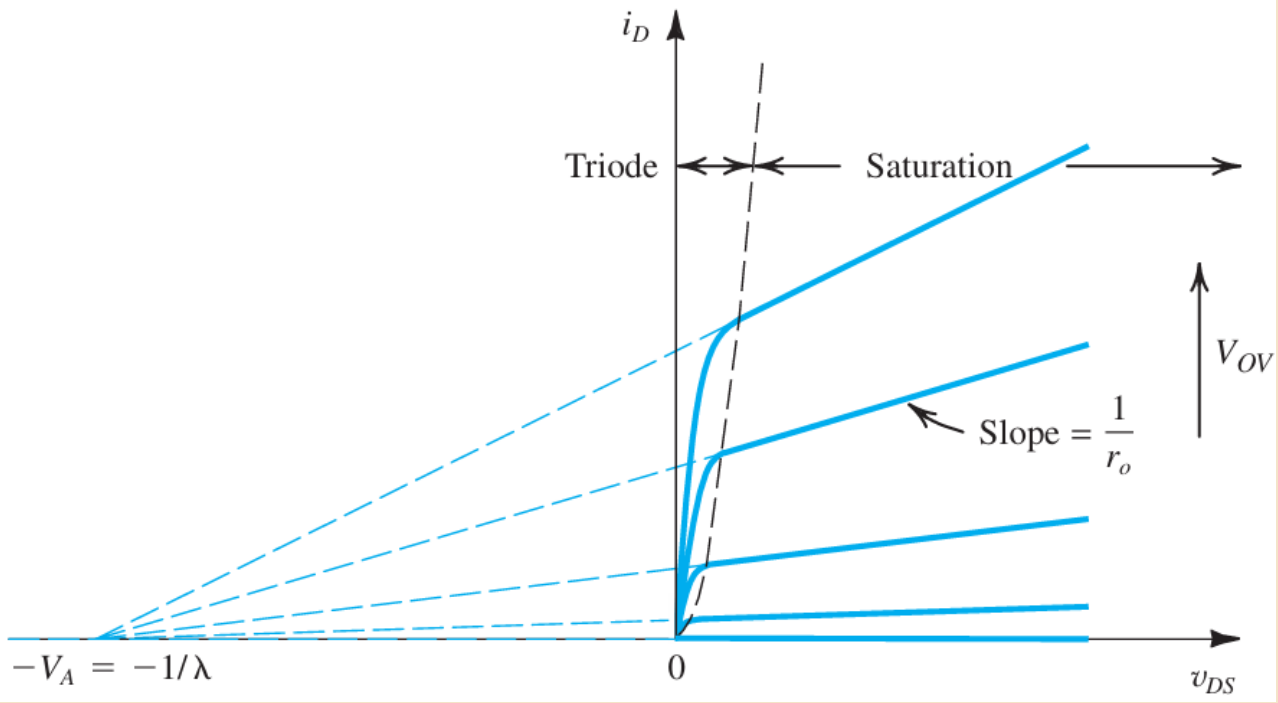


Fig. Effect of v_{DS} on i_D in the saturation region

If these characteristics are extrapolated (extended), they intercept the v_{DS} – axis at the point v_{DS} = $-V_A$ (= $-1/\lambda$), however, i_D = 0. Therefore, $V_A = -1/\lambda$.

Thus, V_A is a process-technology parameter with the dimensions of V. for a given process, V_A is proportional to channel length and is given by

$$V_A = V_A' L$$

where V_A' is entirely process-technology dependent with the dimensions of V/μm. Typically, V_A' falls in the range of 5 V/μm to 50 V/μm. The voltage V_A is usually referred to as the **Early voltage**.

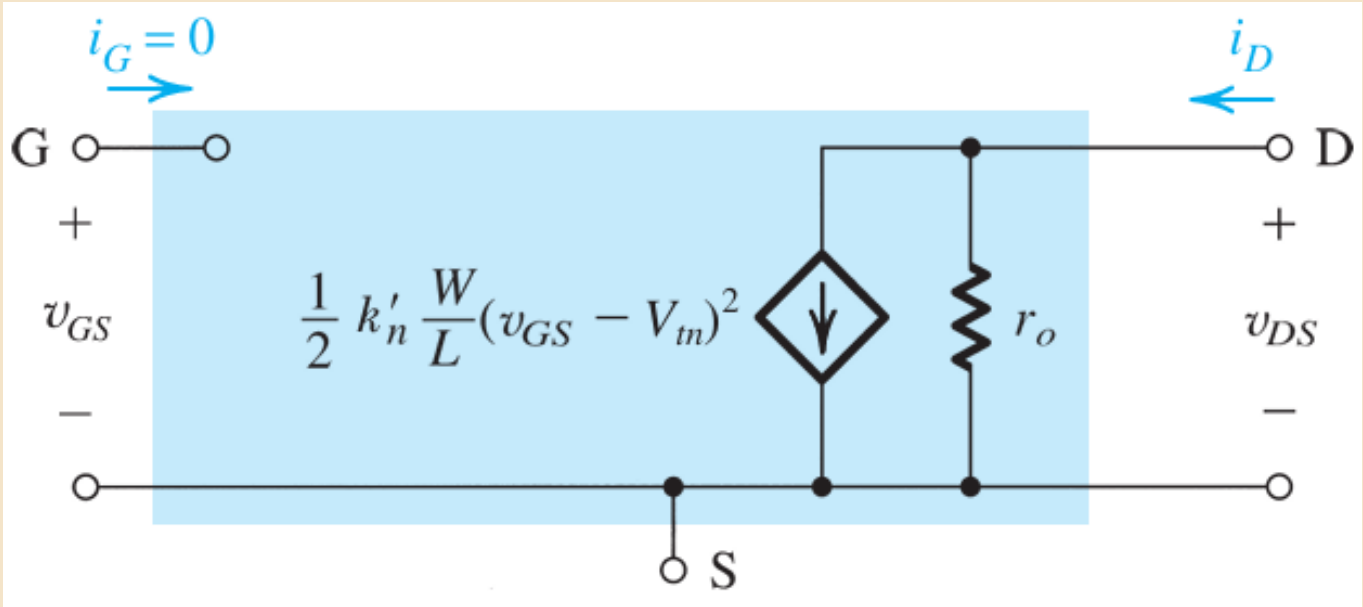
$$r_o \equiv \left[\frac{\partial i_D}{\partial v_{DS}}\right]^{-1}_{v_{GS}\ \text{constant}}$$

$$r_o = \left[\lambda \frac{k_n'}{2}\frac{W}{L}(V_{GS} - V_t)^2\right]^{-1}$$

$$r_o = \frac{1}{\lambda I_D}$$

$$r_o = \frac{V_A}{I_D}$$

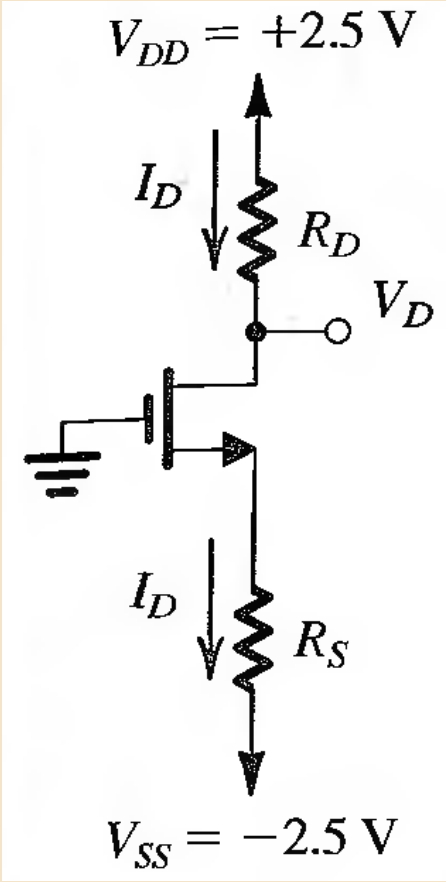
where I_D is the drain current without channel-length modulation taken into account. Thus the output resistance is inversely proportional to the drain current. Now, the large-signal equivalent circuit model incorporating r_o becomes



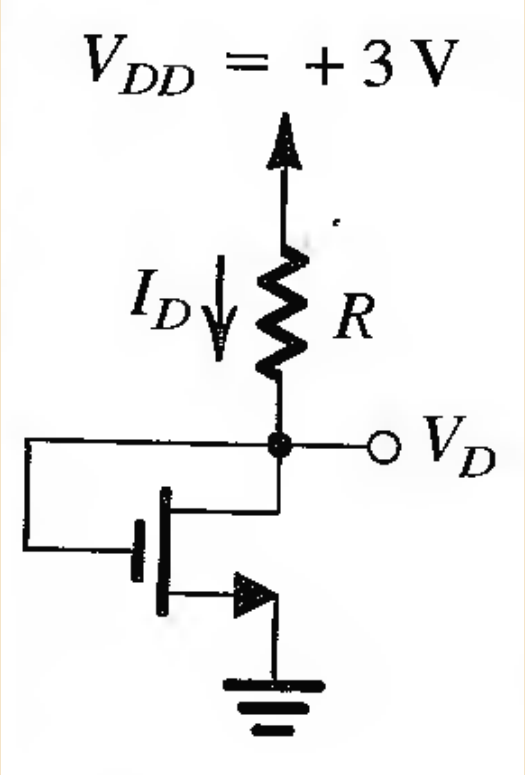
MOSFET CIRCUITS AT DC

Example 1:

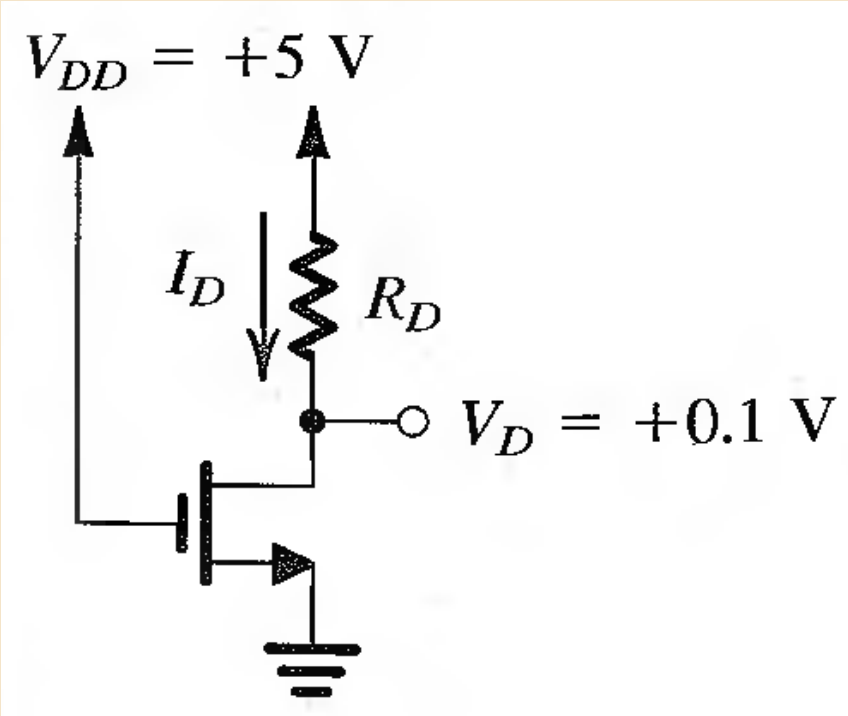
Design the circuit so that the transistor operates at I_D = 0.4 mA and V_D =+0.5 V. The NMOS transistor has V_t, = 0.7 V, $\mu_n C_{ox}$ = 100 μA/V^2 , L = 1μm, and W = 32 μm. Neglect the channel-length modulation effect (i.e., assume that λ = 0).



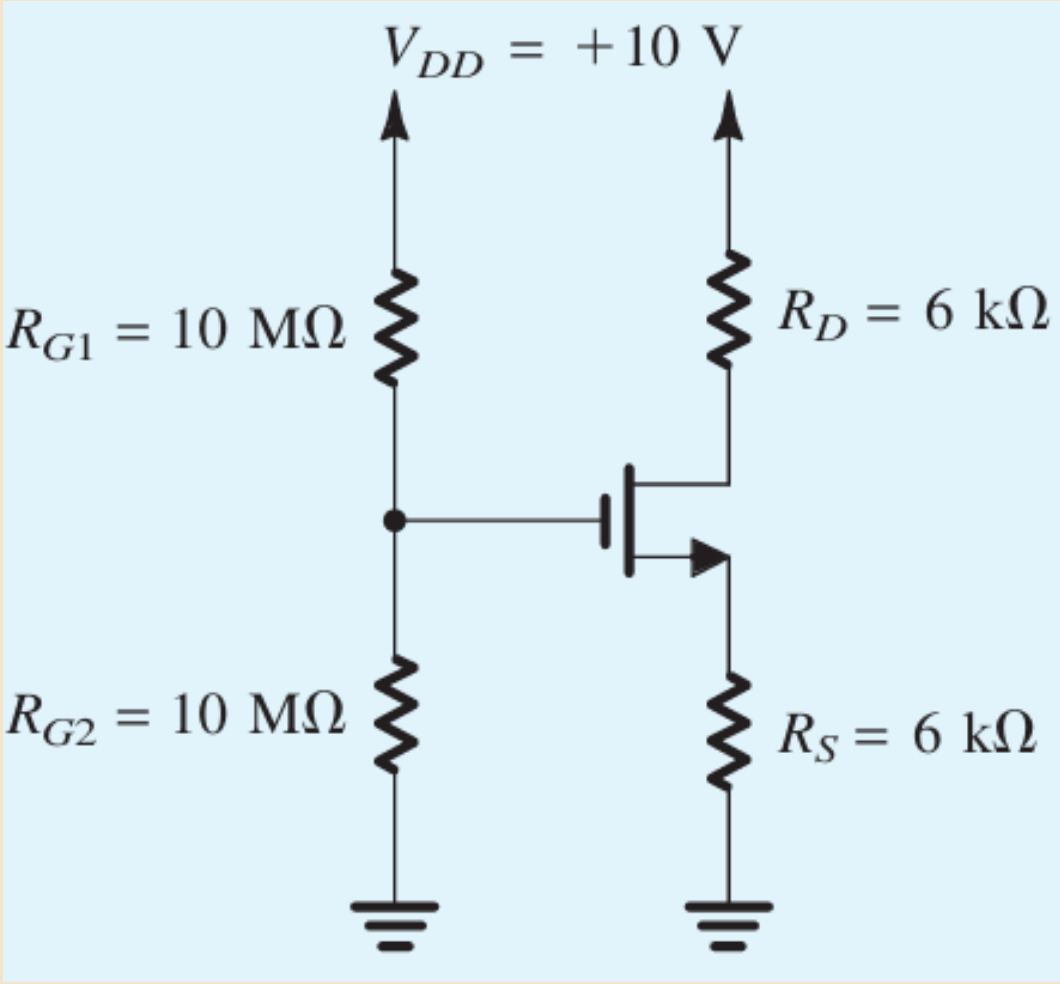
Example 2: Design the circuit to obtain a current I_D of 80 μA. Find the value required for R, and find the dc voltage V_D. Let the NMOS transistor have V_t = 0.6 V, $\mu_n C_{ox}$ = 200 μA/V^2, L = 0.8μm, and W == 4 μm. Neglect the channel-length modulation effect (i.e., assume λ = 0).



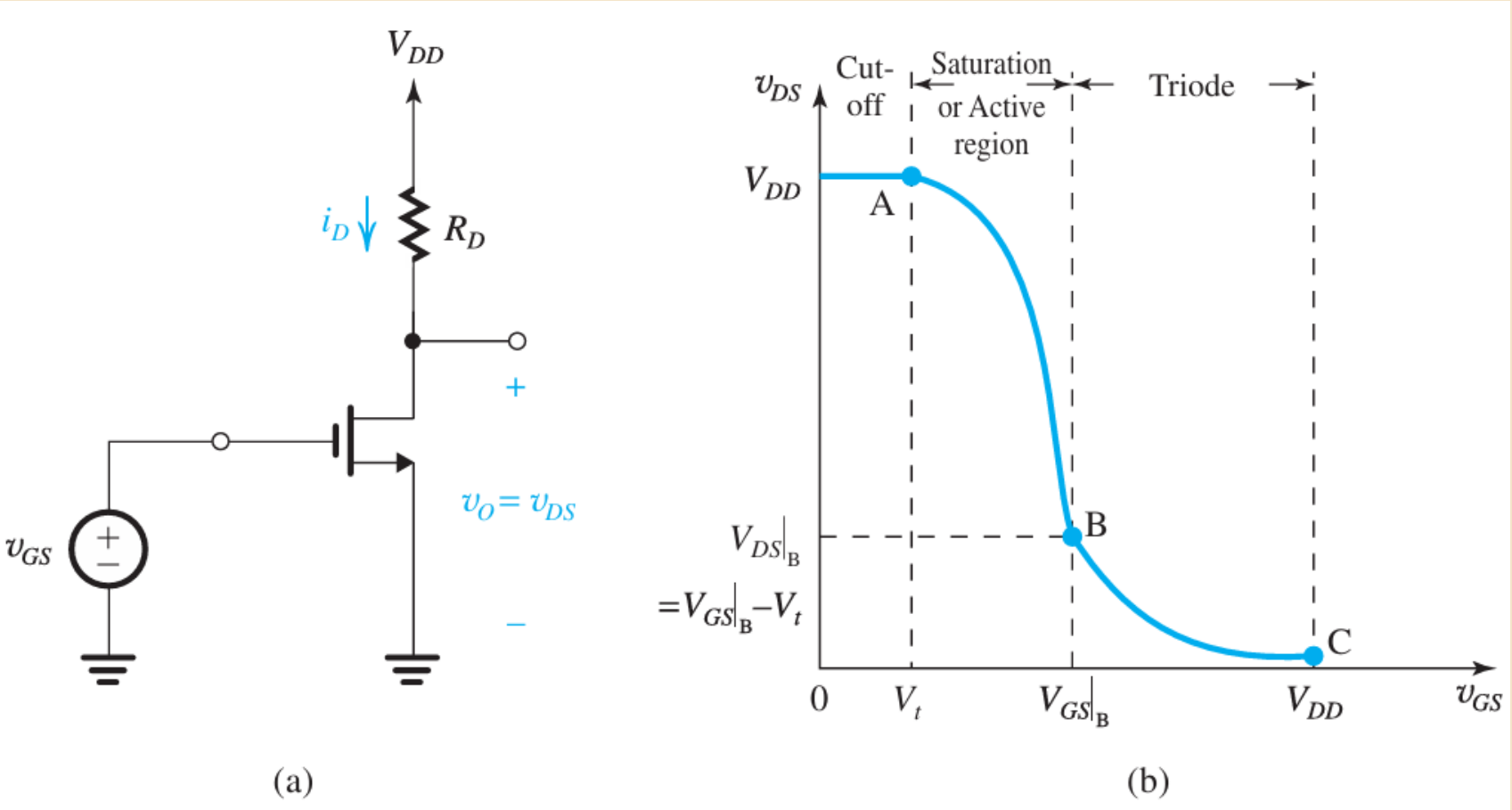
Example 3: Design the circuit to establish a drain voltage of 0.1 V. What is the effective resistance between drain and source at this operating point? Let $V_t = 1$ V and $k_n'(W / L) = 1$ mA/V^2.



Example 4: Analyze the circuit shown in Fig. (a) to determine the voltages at all nodes and the currents through all branchs. Let V_t = 1 V and $k_n'(W / L)$ = 1 mA/V^2. Neglect the channel-length modulation effect (i.e., assume $\lambda = 0$).



MOSFET as an Amplifier



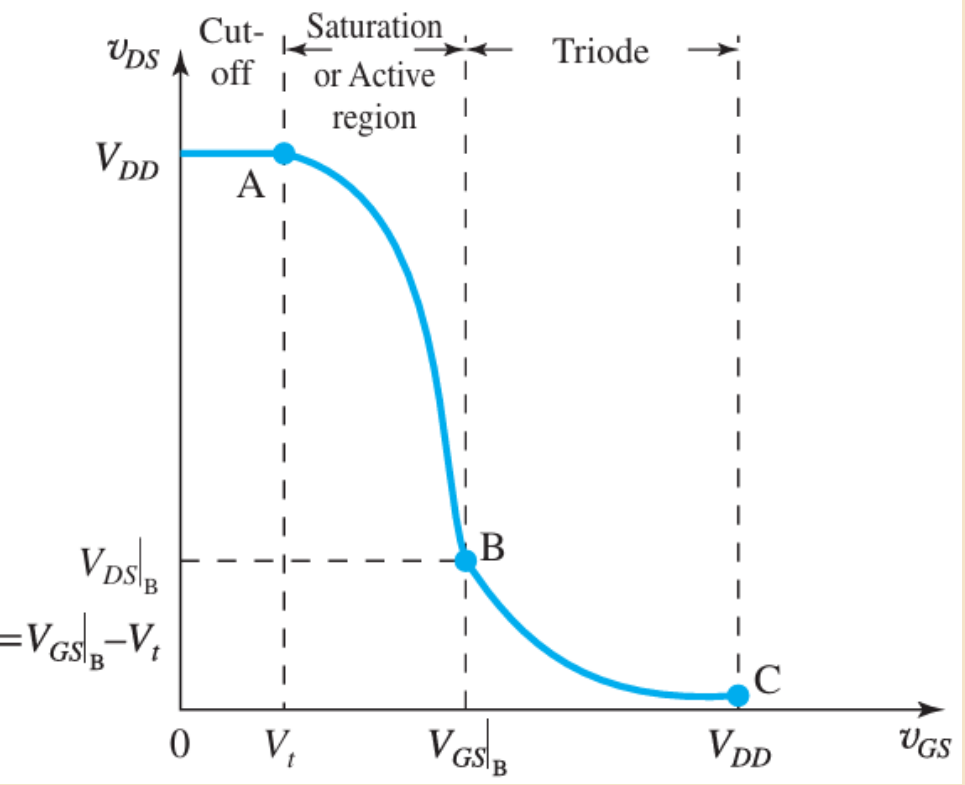
Here v_{GS} is the input voltage, R_D (known as a load resistance) converts the drain current i_D to a voltage ($i_D R_D$), and V_{DD} is the supply voltage that powers up the amplifier.

The output voltage is taken between the drain and ground

The output voltage v_{DS} is given by

$$v_{DS} = V_{DD} - i_D R_D$$

Fig. (a) An NMOS amplifier and (b) its Voltage Transfer Characteristic



$$v_{DS} = V_{DD} - i_D R_D$$

Since $v_{GS} = v_I$, when $v_I < V_t$, the transistor will be cut-off, i_D becomes zero, and $v_O = v_{DS} = V_{DD}$). Operation will be at the point labeled A. As v_I exceeds V_t, the transistor turns on, i_D increases, and from above equ, v_O (= v_{DS}) decreases. Since initially as v_O is high, the transistor will be operating in the saturation region or active region. This continues as v_{GS} is increased until the value of v_{GS} is reached that results in v_{DS} becoming lower than v_{GS} by V_t volts [point B on the VTC in Fig.(b)]. For v_{GS} greater than that at point B, the transistor operates in the triode region and v*DS* decreases more slowly.

When a MOSFET is operated as an amplifier, its operating point is confined to the segment AB at all times.
An expression for the segment AB is given as

$$v_{DS} = V_{DD} - \frac{1}{2} k_n R_D (v_{GS} - V_t)^2$$

This is obviously a nonlinear relationship. Linear (or almost-linear) amplification can be obtained by using the technique of biasing the MOSFET.

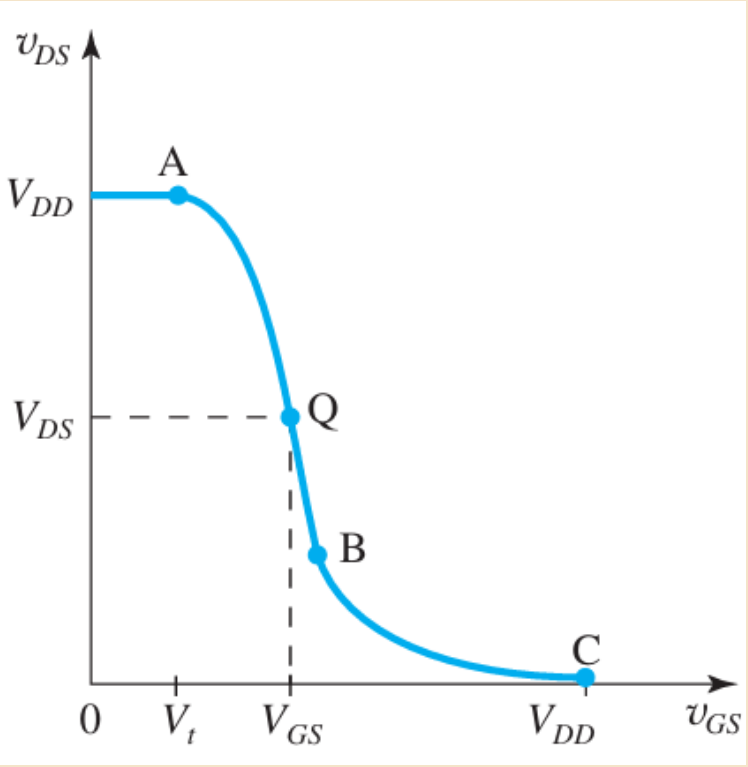


Fig. Biasing the MOSFET amplifier at a point Q located on the segment AB of the VTC

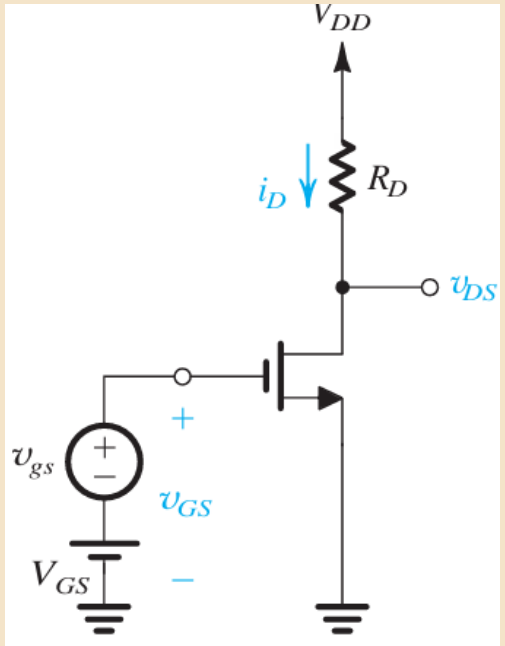
Biasing enables us to obtain almost-linear amplification from the MOSFET. A dc voltage VGS is selected to obtain operation at a point Q on the segment AB of the VTC. The coordinates of Q are the dc voltages VGS and VDS , which are related by

$$V_{DS} = V_{DD} - \frac{1}{2} k_n R_D (V_{GS} - V_t)^2$$

Point **Q** is known as the bias point or the **dc operating point**. Also, since at **Q** no signal component is present, it is also known as the **quiescent point**.

The signal to be amplified, v_{gs}, a function of time t, is superimposed on the bias voltage V_{GS}, as shown in Fig. (a). Thus the total instantaneous value of V_{GS} becomes $v_{GS}(t) = V_{GS} + v_{gs}(t)$

The amplitude of vgs is small enough to restrict the excursion of the instantaneous operating point to a short, almost-linear segment of the VTC around the bias point Q. This is the essence of obtaining linear amplification from the nonlinear MOSFET.



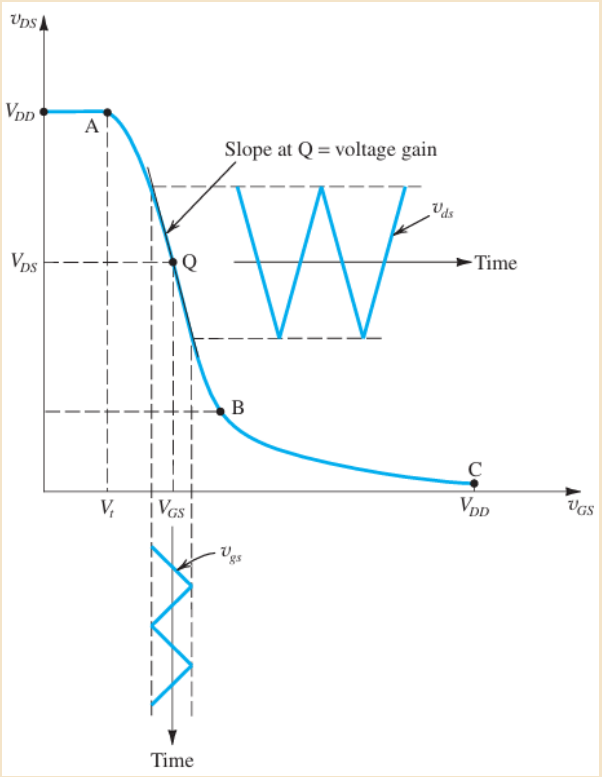
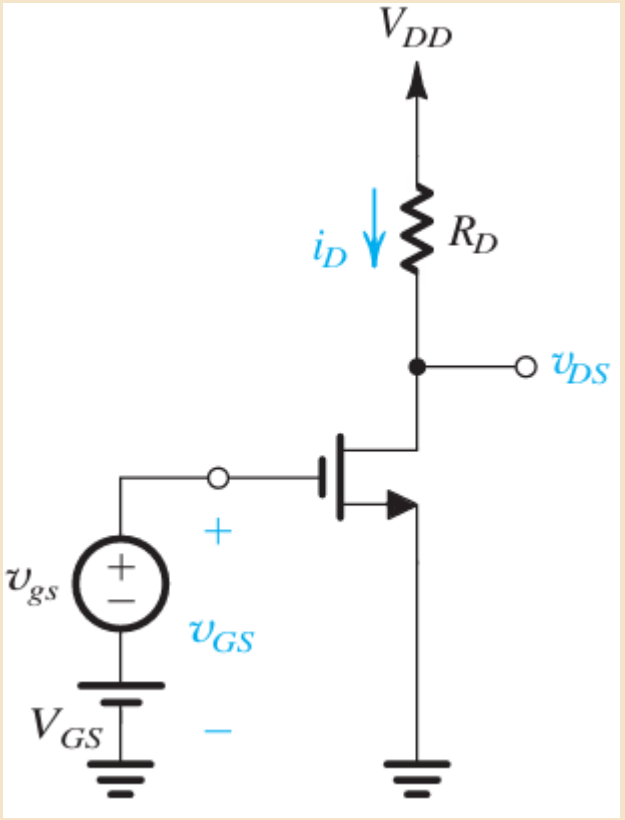
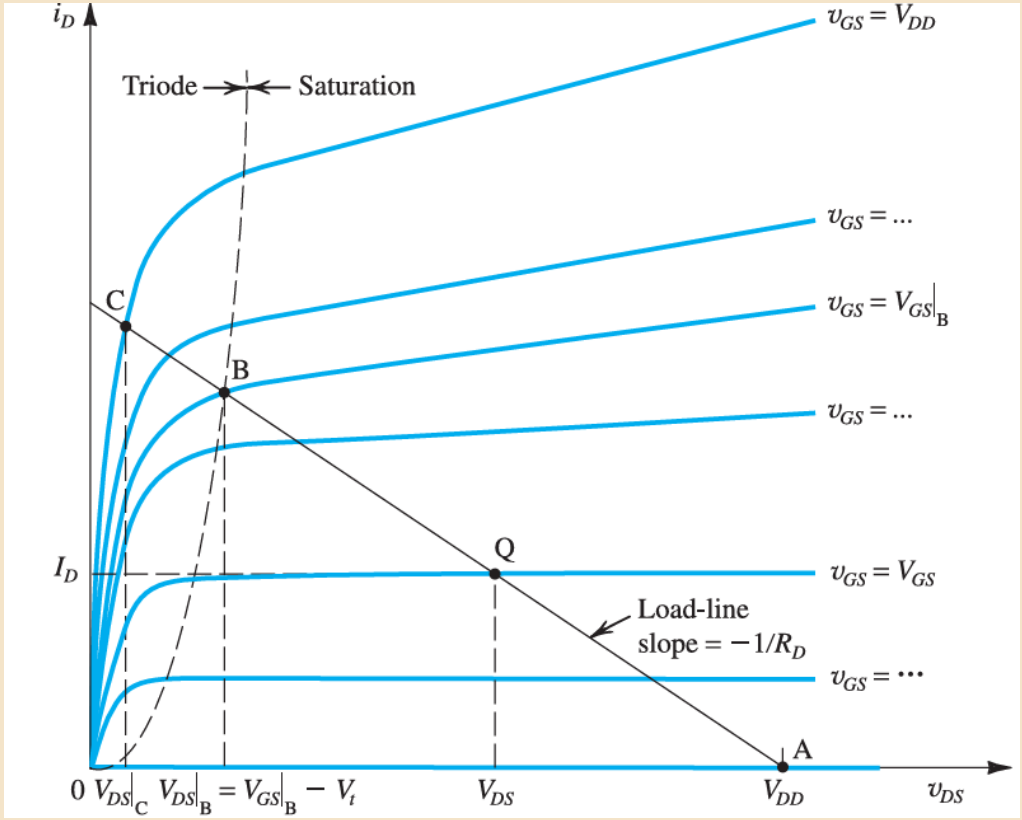
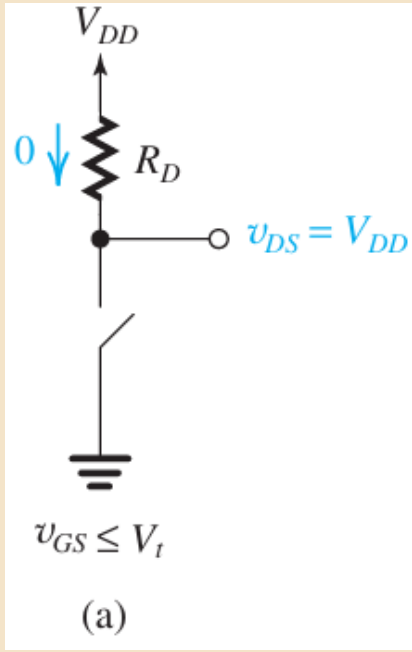


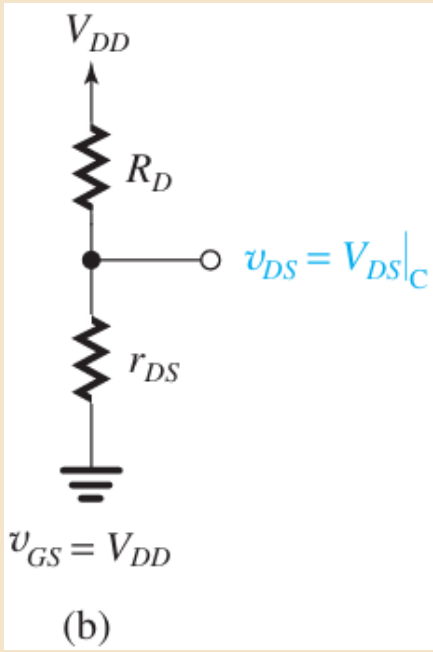
Fig. The MOSFET amplifier with a small time-varying signal $v_{gs}(t)$ superimposed on the dc bias voltage V_{GS}. The MOSFET operates on a short almost-linear segment of the VTC around the bias point Q and provides an output voltage $v_{ds} = A_v v_{gs}$.

Determining the VTC by Graphical Analysis



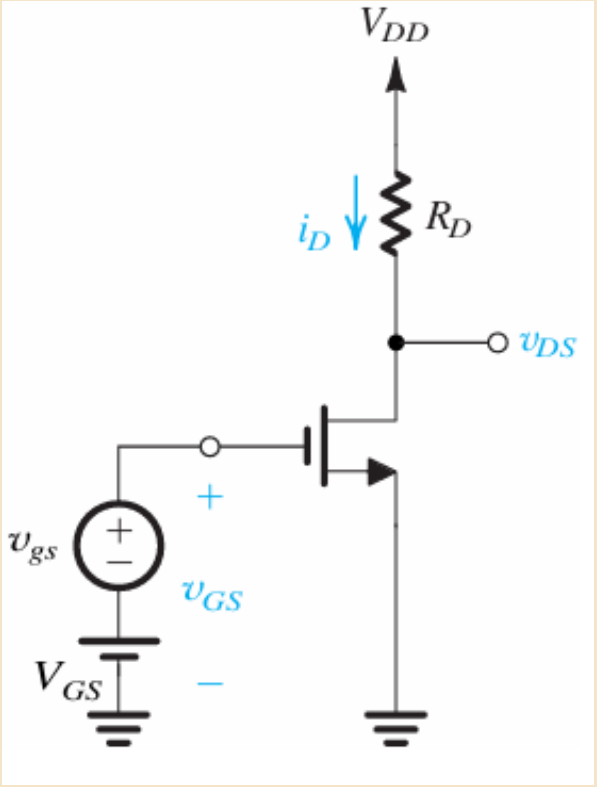






Operation of the MOSFET as a switch:
(a) open, corresponding to point A
(b) closed, corresponding to point C

Small-Signal Operation and Models



Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

$$V_{DS} = V_{DD} - R_D I_D$$

To ensure saturation-region operation, we must have

$$V_{DS} > V_{OV}$$

$$v_{GS} = V_{GS} + v_{gs}$$

resulting in a total instantaneous drain current i_D,

$$i_d = k_n (V_{GS} - V_t) v_{gs} \cdot V_t)^2$$

$$= \frac{1}{2} k_n (V_{GS} - V_t)^2 + k_n (V_{GS} - V_t) v_{gs} + \frac{1}{2} k_n v_{gs}^2$$

$$\frac{1}{2} k_n v_{gs}^2 \ll k_n (V_{GS} - V_t) v_{gs}$$

$$v_{gs} \ll 2(V_{GS} - V_t)$$

$$v_{gs} \ll 2V_{OV}$$

If this small-signal condition is satisfied, we may neglect the last term in Eq. and express i_D as

$$i_D \simeq I_D + i_d$$

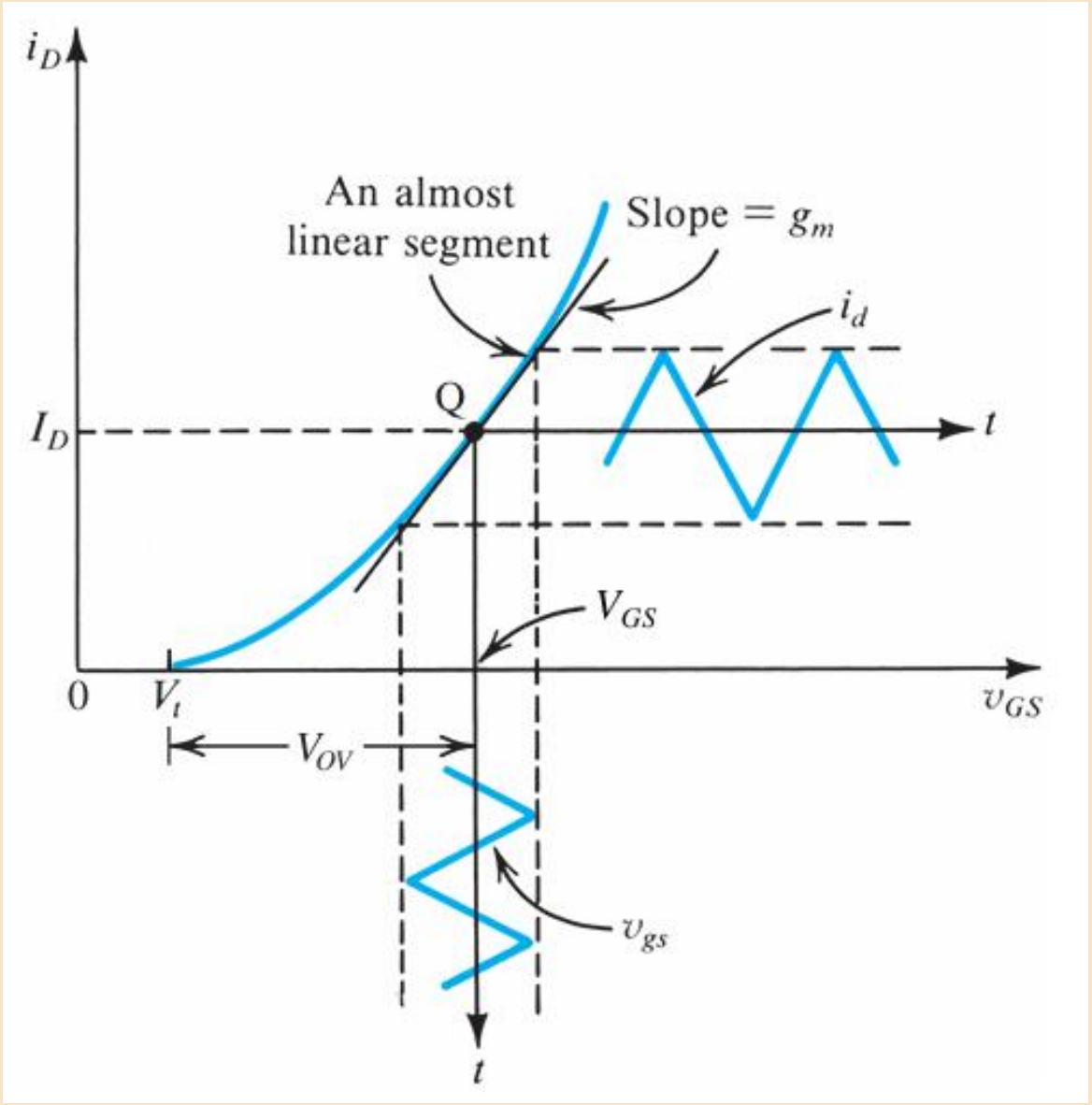
$$i_d = k_n (V_{GS} - V_t) v_{gs}$$

The parameter that relates i_d and v_{gs} is the MOSFET **transconductance** g_m,

$$g_m \equiv \frac{i_d}{v_{gs}} = k_n (V_{GS} - V_t)$$

$$g_m = k_n V_{OV}$$

$$g_m \equiv \left.\frac{\partial i_D}{\partial v_{GS}}\right|_{v_{GS}=V_{GS}}$$



Small-signal operation of the MOSFET amplifier.

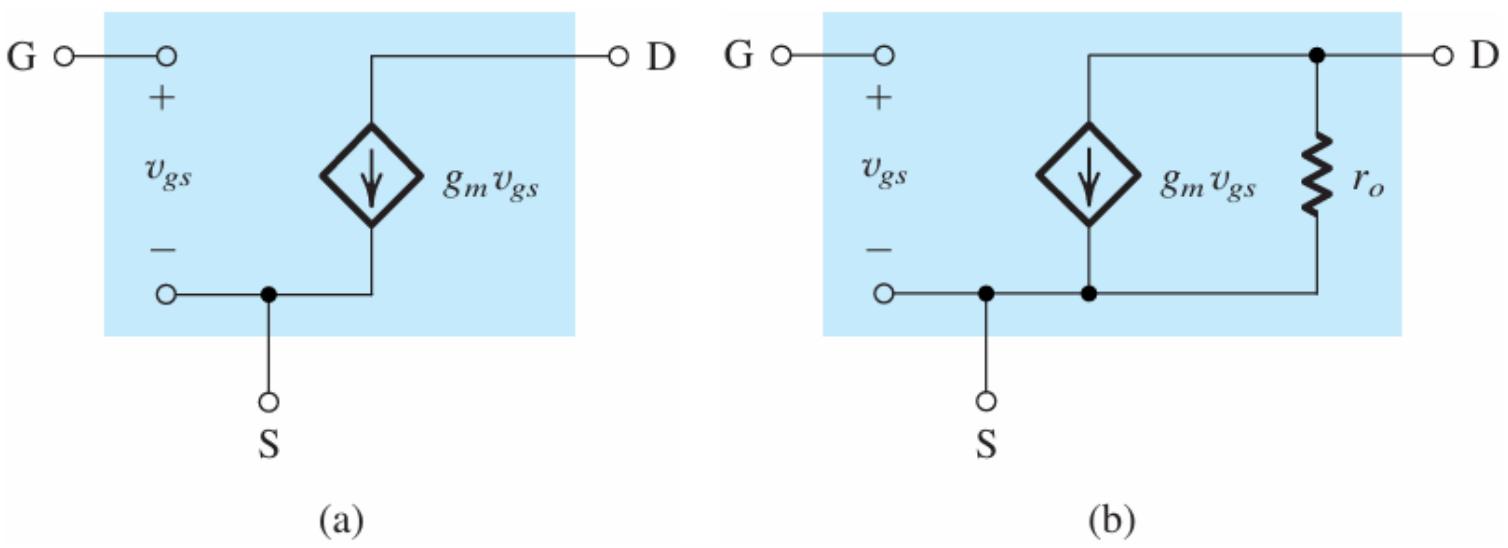
$$v_{DS} = V_{DD} - R_D i_D$$

$$v_{DS} = V_{DD} - R_D(I_D + i_d)$$

$$v_{ds} = -i_d R_D = -g_m v_{gs} R_D$$

$$A_v \equiv \frac{v_{ds}}{v_{gs}} = -g_m R_D$$

Small-Signal Equivalent-Circuit Models



Small-signal models for the MOSFET: (a) neglecting the dependence of i_D on v_{DS} in saturation (the channel-length modulation effect); and (b) including the effect of channel-length modulation, modeled by output resistance $r_o = V_A / I_D$

Typically, r_o is in the range of 10 kΩ to 1000 kΩ. It follows that the accuracy of the small signal model can be improved by including ro in parallel with the controlled source, as shown in Fig. (b).

Considering the finite output resistance r_o, the voltage-gain can be expressed as: $$A_v = \frac{v_{ds}}{v_{gs}} = -g_m(R_D \| r_o)$$

Thus, the finite output resistance r_o results in a reduction in the magnitude of the voltage gain.

The Transconductance g_m

We know that

$$g_m \equiv \frac{i_d}{v_{gs}} = k_n(V_{GS} - V_t) \qquad g_m \equiv \frac{i_d}{v_{gs}} = k_n(V_{GS} - V_t) \qquad = k_n V_{OV}$$

$$k_n = k_n' \ (W/L)$$

Therefore $g_m = k_n'(W/L)(V_{GS} - V_t) = k_n'(W/L)V_{OV}$

This relationship indicates that g_m is proportional to the process transconductance parameter $k_n' = \mu_n C_{ox}$ and to the W/L ratio of the MOS transistor; hence to obtain relatively large transconductance the device must be short and wide.
It is also observed that for a given device the transconductance is proportional to the overdrive voltage, $V_{OV} = V_{GS} - V_t$.

$$I_D = \frac{1}{2}k_n(V_{GS} - V_t)^2 = \frac{1}{2}k_n V_{OV}^2 \quad \Rightarrow \quad k_n' \frac{W}{L} = \frac{2I_D}{V_{OV}^2} \qquad V_{OV} = \sqrt{\frac{2I_D}{k_n' \frac{W}{L}}} = \sqrt{\frac{2I_D}{k_n}}$$

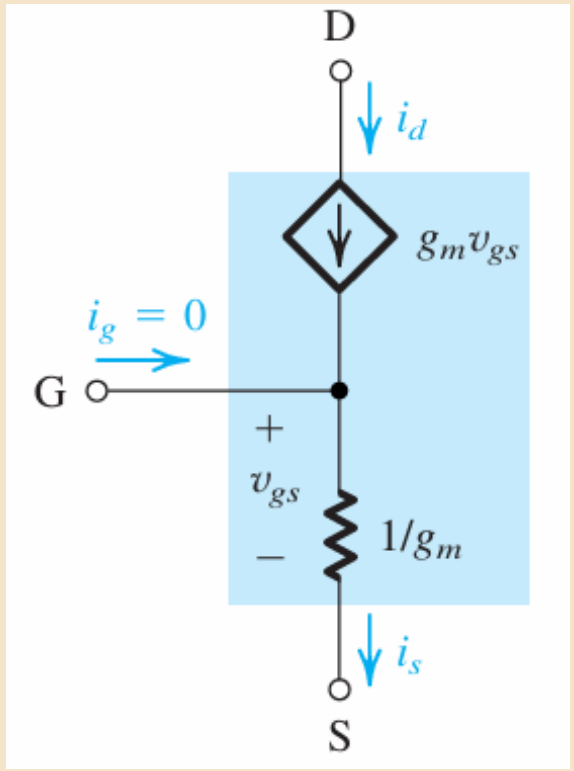
Therefore $g_m = k_n' \frac{W}{L}\sqrt{\frac{2I_D}{k_n' \frac{W}{L}}} = \sqrt{2I_D k_n' \frac{W}{L}} = \sqrt{2k_n'}\sqrt{\frac{W}{L}}\sqrt{I_D}$

For a given MOSFET,

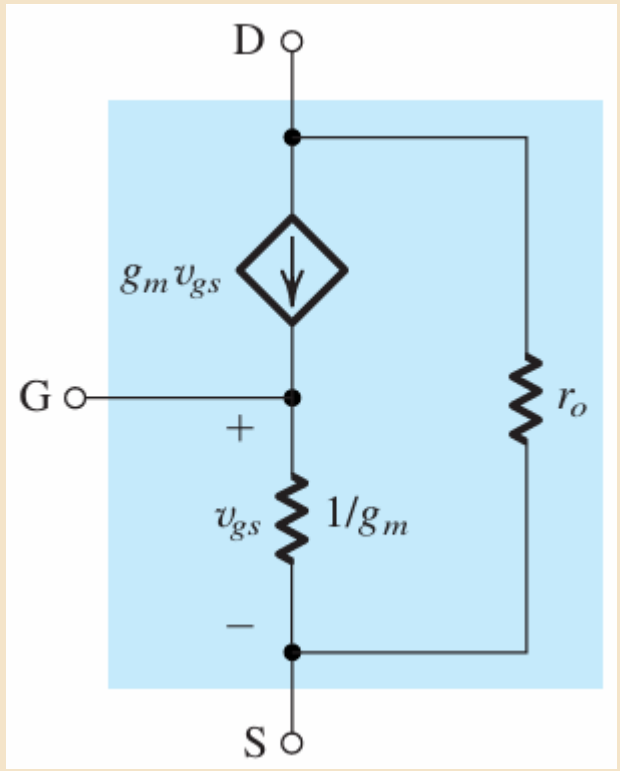
1. g_m is proportional to the square root of the dc bias current.
2. At a given bias current, g_m is proportional to $\sqrt{\frac{W}{L}}$.

(OR) $g_m = \sqrt{2I_D k_n' \frac{W}{L}} = \sqrt{2I_D \frac{2I_D}{V_{OV}^2}} = \frac{2I_D}{v_{ov}}$

The T Equivalent-Circuit Model

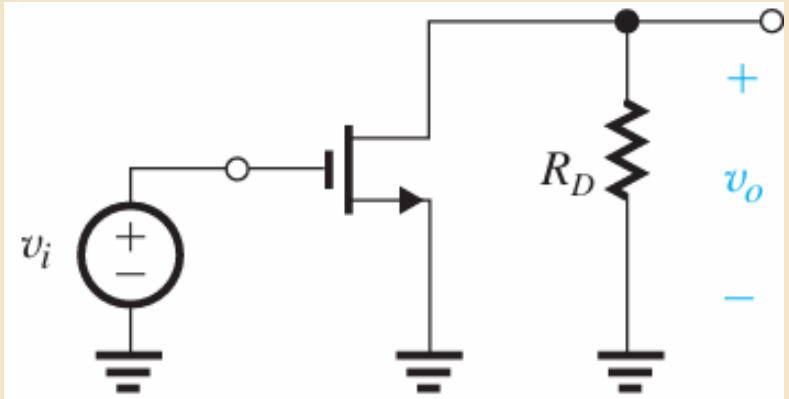


T equivalent-circuit model for the MOSFET

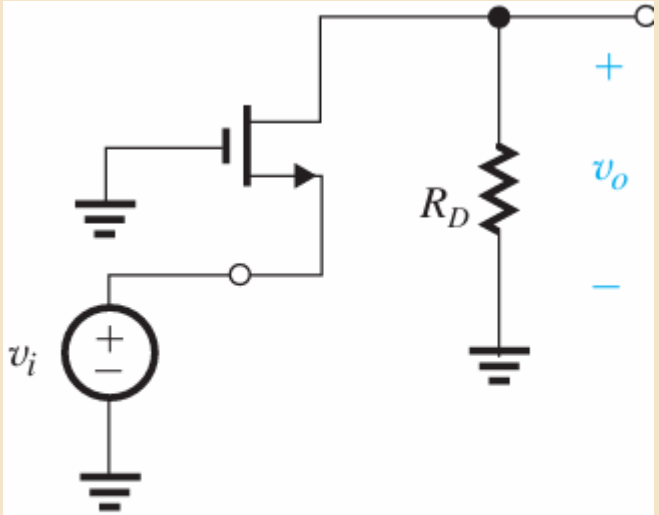


T model of the MOSFET augmented with the drain-to-source resistance r_o

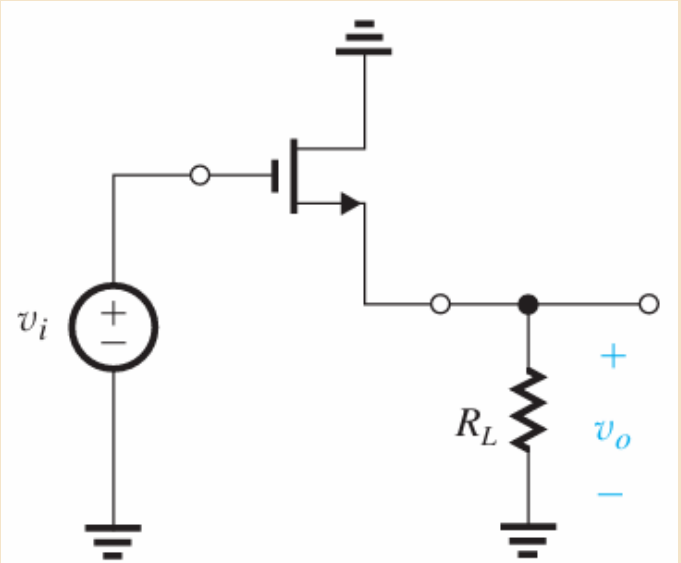
Basic MOSFET Amplifier Configurations



(a) Common Source (CS)



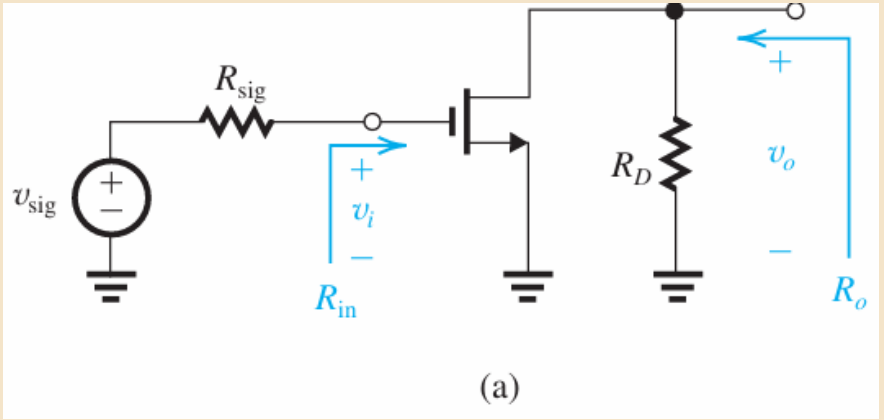
(b) Common Gate (CG)



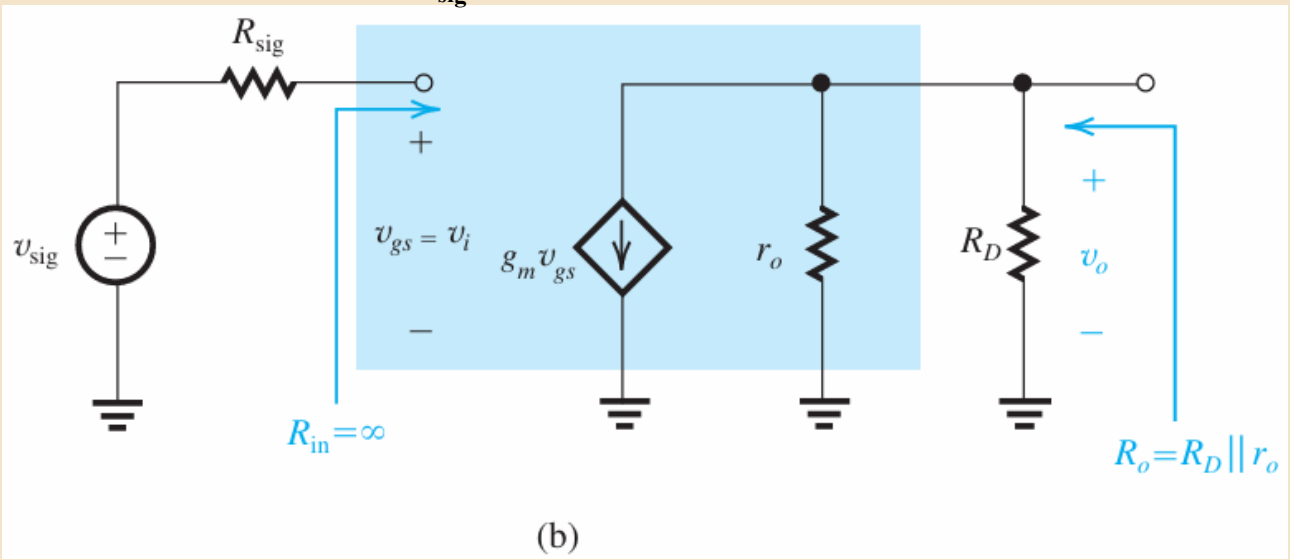
(c) Common Drain (CD)

The Common-Source (CS) Amplifier

Of the three basic MOS amplifier configurations, the common source is the most widely used. Typically, in an amplifier formed by cascading a number of stages, the bulk of the voltage gain is obtained by using one or more common-source stages in the cascade.



Common-source amplifier fed with a signal v_{sig} from a generator with a resistance R_{sig} . The bias circuit is omitted



The common-source amplifier with the MOSFET replaced with its hybrid π model

Characteristic parameters: R_{in}, Av_0 and R_0:

The **input resistance** is obviously infinite, $R_{in} = \infty$

The output voltage v_0 is found by multiplying the current by the total resistance between the output node and ground,

$$v_o = -(g_m v_{gs})(R_D \| r_o)$$

Since $v_{gs} = v_i$, the open-circuit voltage gain $A_{vo} \equiv v_o / v_i$ can be obtained as

$$A_{vo} = -g_m(R_D \| r_o)$$

The transistor output resistance reduces the magnitude of the voltage gain. R_D is usually much lower than and the effect of on reducing $|Av_0|$ is slight (less than 10% or so). Thus in many cases we can neglect r_o and express Av_0 simply as

$$A_{vo} \simeq (-g_m R_D)$$

The **output resistance** $R_o = R_D \| r_o$

By neglecting r_o, $R_o \simeq R_D$

Observations from the analysis of CS Amplifier:

(i) The input resistance is ideally infinite.

(ii) The output resistance is moderate to high (in the kilohms to tens of kilohms range). Reducing R_D to lower R_0 is not a viable proposition, since the voltage gain is also reduced. Alternatively, if a low output resistance (in the ohms to tens of ohms range) is needed, a source follower or Common Drain stage can be used.

(iii) The open-circuit voltage gain Av_0 can be high, making the CS configuration the work horse in MOS amplifier design.

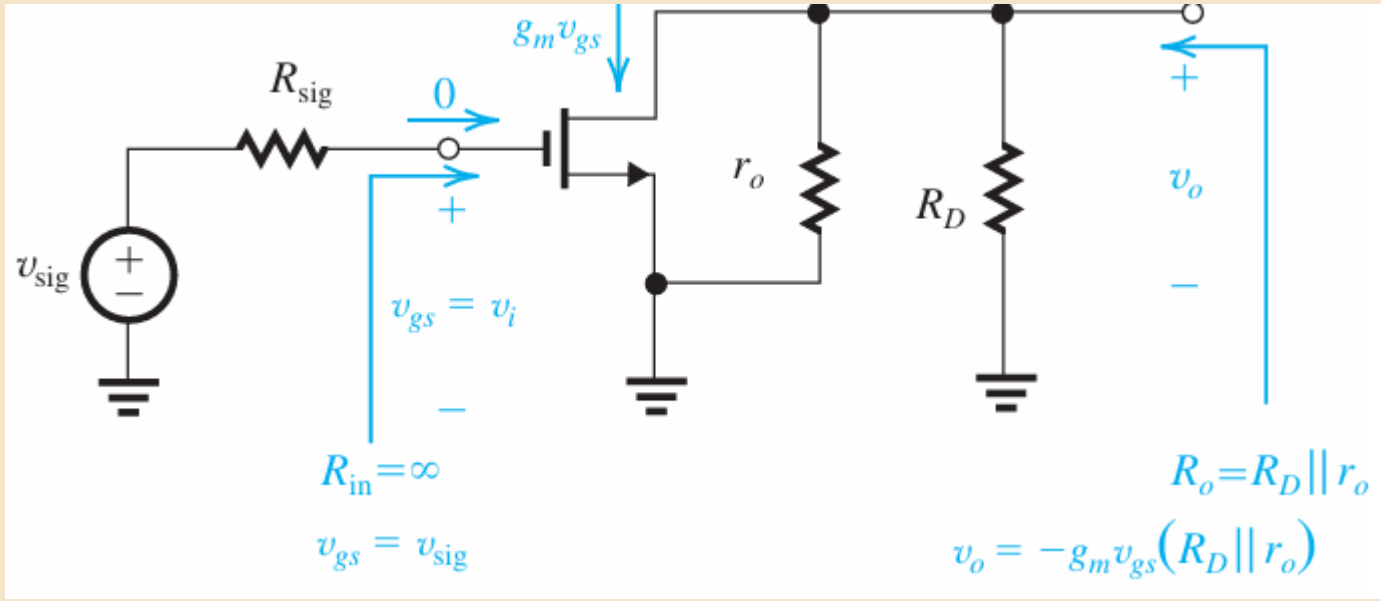
Overall Voltage Gain: $v_i = v_{sig}$

If a load resistance **R_L** is connected to the output terminal of the amplifier, this resistance will appear in parallel with **R_D**

$$A_v = -g_m(R_D \| R_L \| r_o)$$

As $v_i = v_{sig}$, the overall voltage gain G_v is same as A_v.

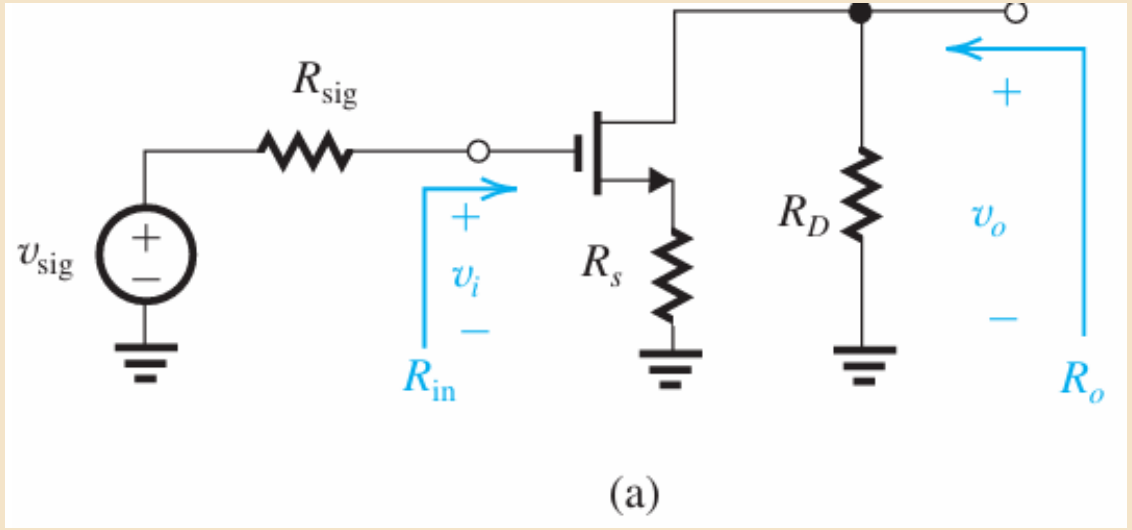
$$G_v = A_v = -g_m(R_D \| R_L \| r_o)$$

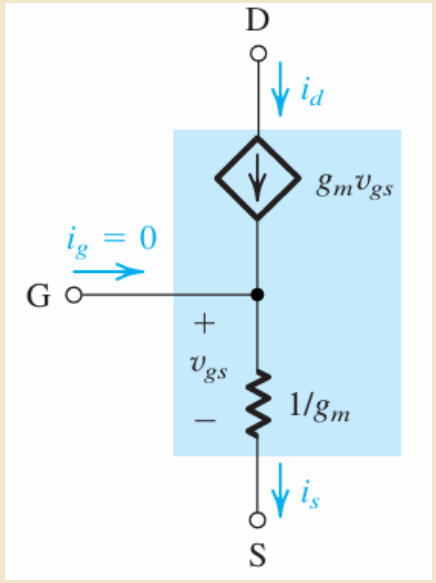


Performing the analysis directly on the circuit diagram with the MOSFET model used implicitly

The Common-Source Amplifier with a Source Resistance

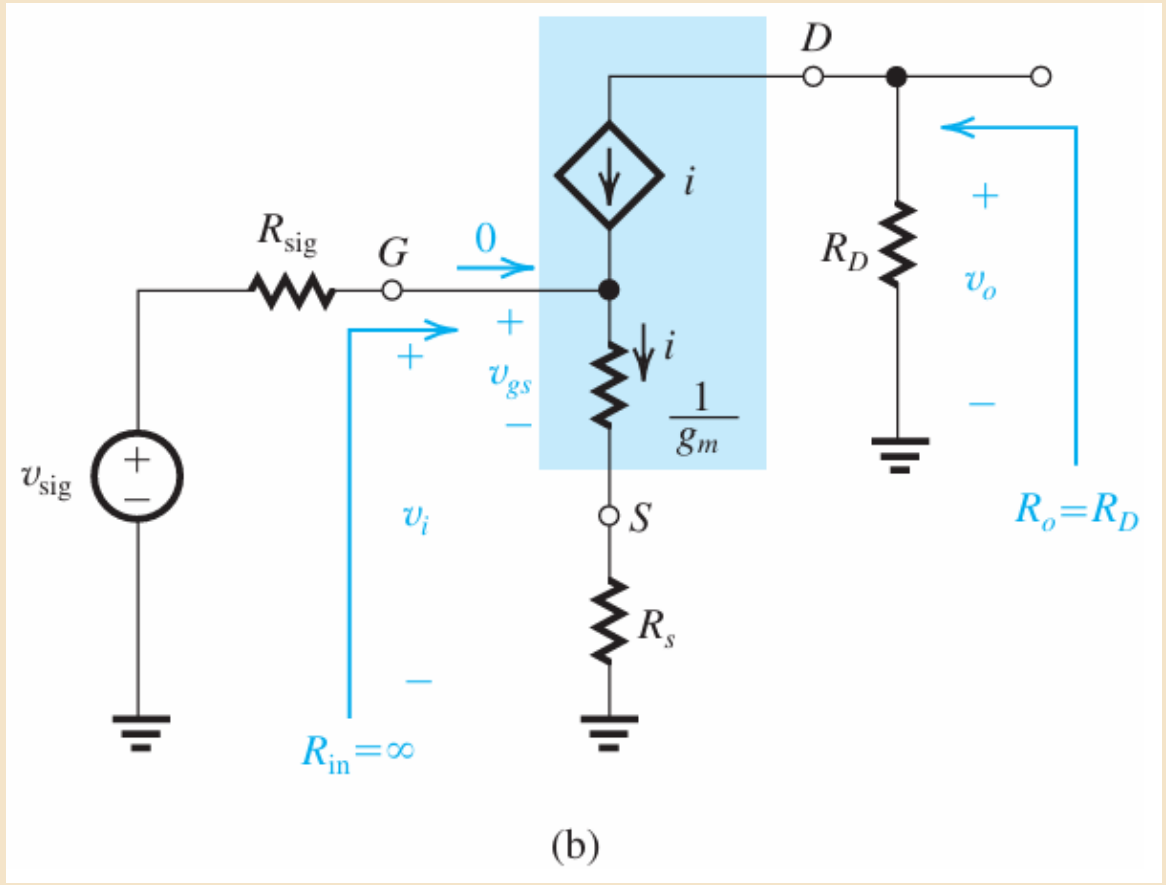
In general, whenever a resistance is connected in the source lead, the T model is preferred.





T model

The CS amplifier with a source resistance Rs : (a) Circuit without bias details; (b) Equivalent circuit with the MOSFET represented by its T model



From Fig. (b) we see that the input resistance R_{in} is infinite and thus $v_i = v_{sig}$. Unlike the CS amplifier, however, here only a fraction of v_i appears between gate and source as v_{gs} . It can be determined from the voltage divider composed of across the amplifier input, as follows:

$$v_{gs} = v_i \frac{1/g_m}{1/g_m + R_s} = \frac{v_i}{1 + g_m R_s}$$

$$v_i = v_{gs}(1 + g_m R_s)$$

The output voltage v_o is obtained by multiplying the controlled-source current i by R_D,

$$v_o = -iR_D \qquad v_o = -g_m v_{gs} R_D$$

$$v_i = v_{gs}(1 + g_m R_s)$$

Thus, the voltage gain Av_0 can be found as $$A_{v0} = \frac{v_o}{v_i} = \frac{-g_m v_{gs} R_D}{v_{gs}(1+g_m R_s)} = \frac{-g_m R_D}{1+g_m R_s} = -\frac{R_D}{\frac{1}{g_m}+R_s}$$

The above equation indicates that including the resistance $\mathbf{R_s}$ reduces the voltage gain by the factor $(1 + g_m R_s)$.

The voltage gain between gate and drain is equal to the ratio of the total resistance in the drain (R_D) to the total resistance in the source ($1/g_m+R_S$),

Output Resistance, $R_0 = R_D$

Considering the Load Resistance, R_L, the Voltage gain A_v can be obtained as $$A_v = -\frac{R_D || R_L}{\frac{1}{g_m} + R_s} = -\frac{g_m(R_D || R_L)}{1 + g_m R_s}$$

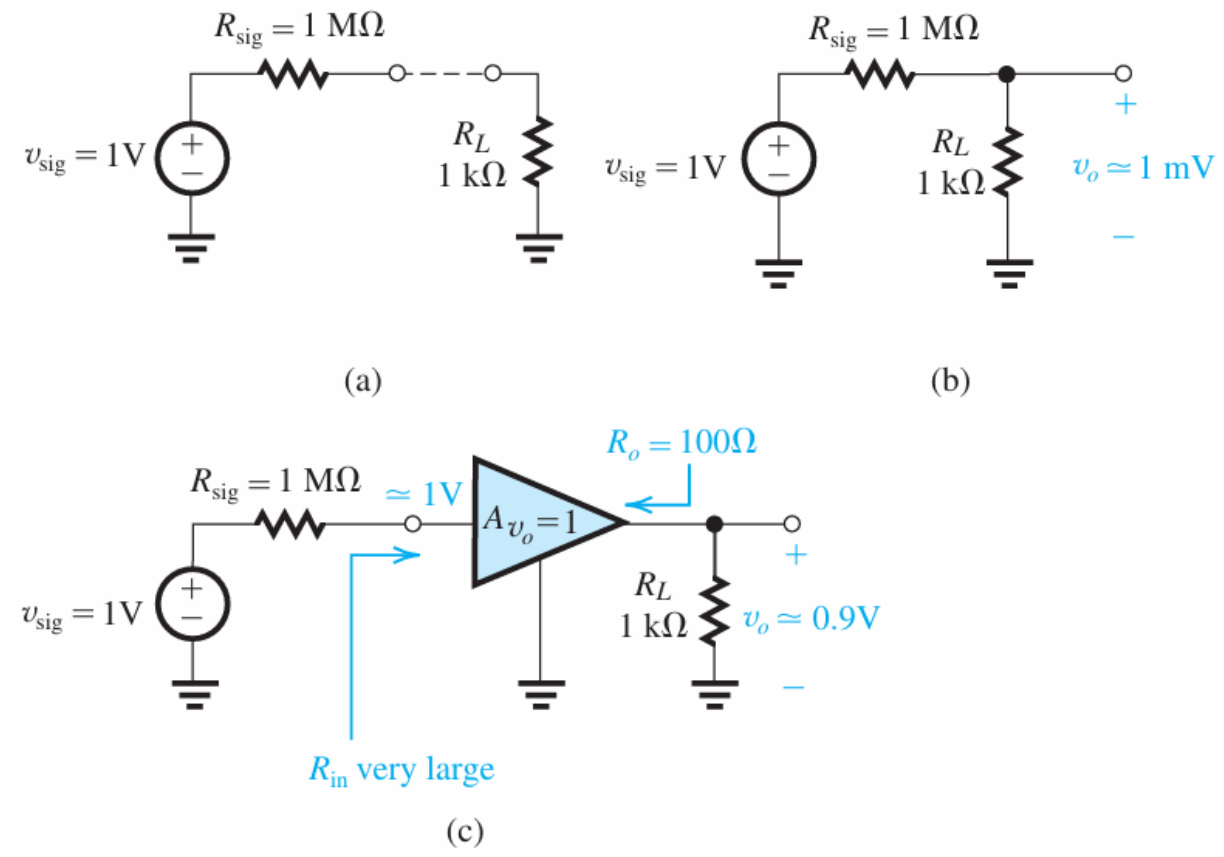
Finally note that because R_{in} is infinite, $v_i = v_{sig}$, and the overall voltage gain G_v is equal to A_v

Including the source resistance increases the bias stability, linearity but reduces the voltage gain.

The Common-Drain (CD) Amplifier or Source Follower

Common Drain Amplifier finds application in the design of both small-signal amplifiers as well as amplifiers that are required to handle large signals and deliver substantial amounts of signal power to a load. The common drain amplifier is more commonly known as the **source follower**.

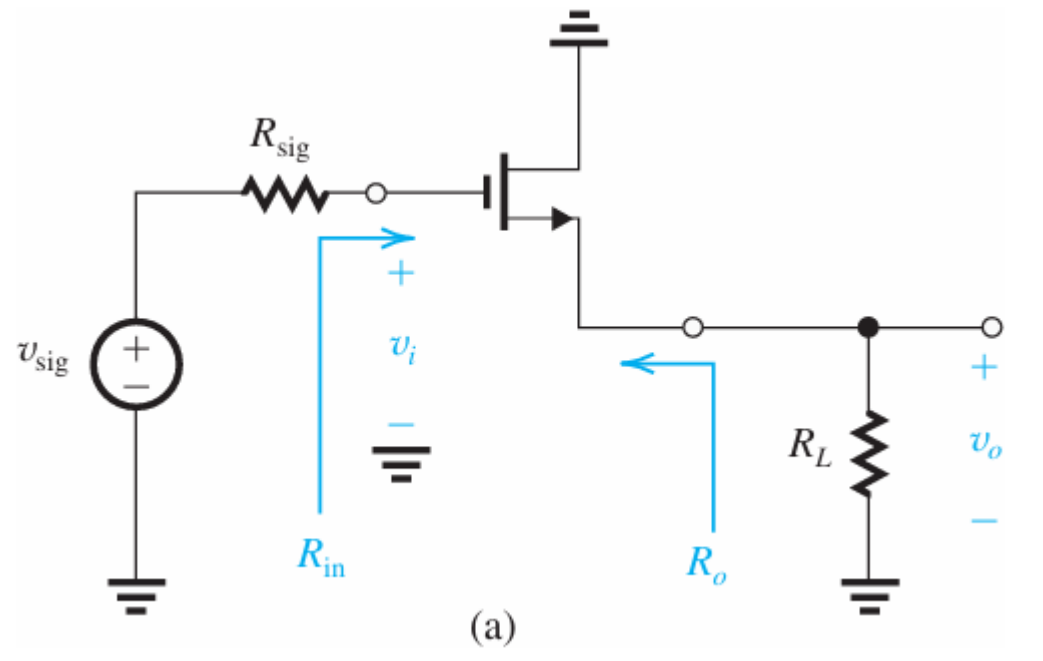
The Need for Voltage Buffers:



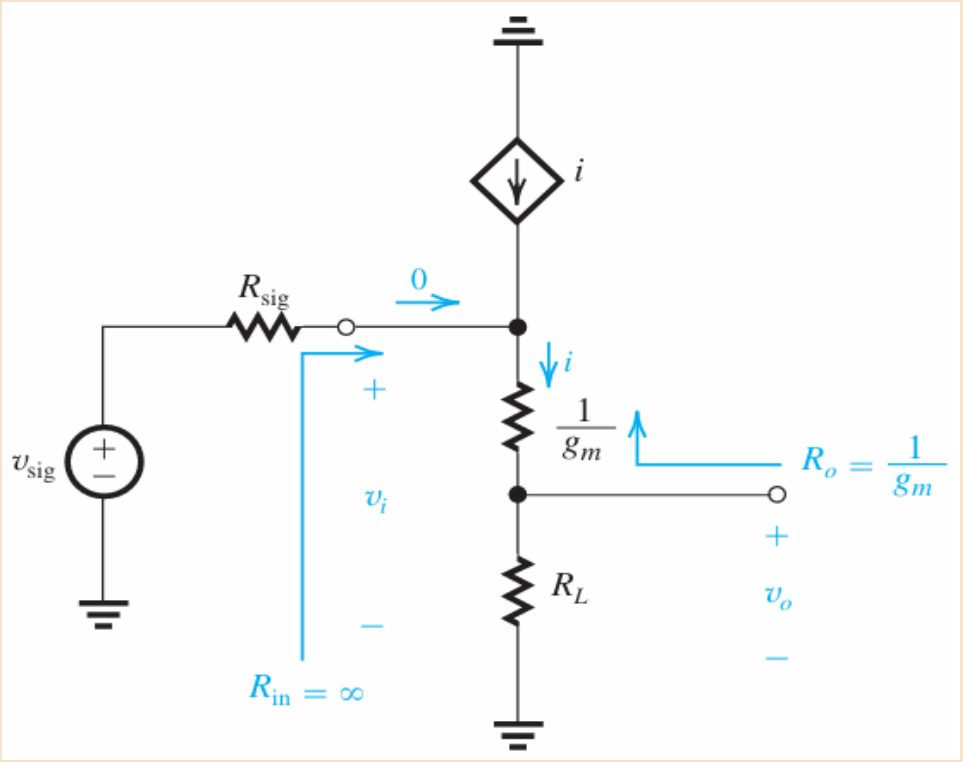
Illustrating the need for a unity-gain buffer amplifier

Since the amplifier has a low output resistance (100), 90% of this signal (0.9 V) will appear at the output, obviously a very significant improvement over the situation without the amplifier.

Analysis of CD Amplifier



Common-drain amplifier or source follower



Equivalent circuit of the source follower obtained by replacing the MOSFET with its T model neglecting r_0.

Input Resistance, $R_{in} = \infty$

Output Resistance, R_0 is found by setting (i.e., by grounding the gate). Now looking back into the output terminal, excluding R_L, we simply see $1/g_m$, thus

$$R_0 = 1/g_m$$

Voltage gain, $A_v = \frac{v_o}{v_i}$

Here, $i = g_m v_{gs}$ From the circuit, $v_{gs} = v_i - v_0$

Therefore, $i = g_m(v_i - v_0) = \frac{v_0}{R_L}$ $g_m v_i = g_m v_0 + \frac{v_0}{R_L} = v_0(\frac{1}{R_L} + g_m)$ $A_v = \frac{v_o}{v_i} = \frac{g_m R_L}{1+g_m R_L} = \frac{R_L}{\frac{1}{g_m}+R_L}$

Setting $R_L = \infty$, $A_{v0} = 1$

The unity open-circuit voltage gain indicates that the voltage at the source terminal will follow that at the input, hence the name source follower.

$$G_v = A_v = \frac{v_o}{v_i} = \frac{g_m R_L}{1 + g_m R_L} = \frac{R_L}{\frac{1}{g_m} + R_L}$$

As $1/g_m$ is usually low, the voltage gain can be close to unity.

The source follower features

- A very high input resistance (ideally, infinite)
- A relatively low output resistance
- An open-circuit voltage gain that is near unity (ideally, unity).

Thus the source follower is ideally suited for implementing the unity-gain voltage buffer. The source follower is also used as the output (i.e., last) stage in a multistage amplifier, where its function is to equip the overall amplifier with a low output resistance, thus enabling it to supply relatively large load currents without loss of gain.

Comparison of the characteristics of MOSFET Amplifiers

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Table 5.4 Characteristics of MOSFET Amplifiers

	Characteristics[a, b]				
Amplifier type	R_{in}	A_{vo}	R_o	A_v	G_v
Common source (Fig. 5.45)	∞	$-g_m R_D$	R_D	$-g_m(R_D \parallel R_L)$	$-g_m(R_D \parallel R_L)$
Common source with R_s (Fig. 5.47)	∞	$-\frac{g_m R_D}{1+g_m R_s}$	R_D	$\frac{-g_m(R_D \parallel R_L)}{1+g_m R_s}$	$-\frac{g_m(R_D \parallel R_L)}{1+g_m R_s}$
				$-\frac{R_D \parallel R_L}{1/g_m + R_s}$	$-\frac{R_D \parallel R_L}{1/g_m + R_s}$
Common gate (Fig. 5.48)	$\frac{1}{g_m}$	$g_m R_D$	R_D	$g_m(R_D \parallel R_L)$	$\frac{R_D \parallel R_L}{R_{sig} + 1/g_m}$
Source follower (Fig. 5.50)	∞	1	$\frac{1}{g_m}$	$\frac{R_L}{R_L + 1/g_m}$	$\frac{R_L}{R_L + 1/g_m}$

1. The CS configuration is the best suited for realizing the bulk of the gain required in an amplifier. Depending on the magnitude of the gain required, either a single stage or a cascade of two or three stages can be used.
2. Including a resistor Rs in the source lead of the CS stage provides a number of performance improvements at the expense of gain reduction.
3. The low input resistance of the CG amplifier makes it useful only in specific applications. As we shall see in Chapter 9, it has a much better high-frequency response than the CS amplifier. This superiority makes it useful as a high-frequency amplifier, especially when combined with the CS circuit.
4. The source follower finds application as a voltage buffer for connecting a high resistance source to a low-resistance load and as the output stage in a multistage amplifier where its purpose is to equip the amplifier with a low output resistance.

Applications of MOSFET Amplifiers

1. Common-Source (CS) Amplifier

Voltage amplification: Provides high voltage gain.

Audio amplifiers: Amplifies weak input signals.

RF amplifiers: Used in receiver and transmitter stages.

Multistage amplifiers: Used as a voltage-gain stage in integrated circuits.

2. Common-Gate (CG) Amplifier

High-frequency amplifiers: Suitable for high-frequency operation.

RF circuits: Used in wideband RF amplifier stages.

Impedance matching: Matches a low-resistance signal source to a higher-resistance load in suitable designs.

Cascode amplifiers: Used as the common-gate stage to improve bandwidth and reduce the Miller effect.

3. Common-Drain (CD) Amplifier

Also called a source follower.

Voltage buffer: Provides a voltage gain close to unity.

Impedance matching: Converts a high input resistance to a low output resistance.

Load driving: Drives low-resistance loads without significantly loading the preceding stage.

Output stages: Used as a buffer in analog integrated circuits.

Biasing in MOS Amplifier Circuits

Biasing is the process of applying suitable DC voltages and currents to a MOSFET to establish a proper operating point (Q-point) before applying the input signal, characterized by a stable and predictable dc drain current I_D and by a dc drain-to-source voltage V_{DS}.

Need for Biasing:

To keep the MOSFET in saturation: Ensures proper amplification.

To avoid distortion: Prevents the MOSFET from entering cutoff or the triode region during the intended signal swing.

To set the required drain current (I_D): Determines the operating point and influences voltage gain.

To obtain maximum undistorted output swing: Helps prevent clipping of the output signal.

To improve Q-point stability: Reduces the effects of variations in temperature and MOSFET parameters.

Biasing Techniques:

Biasing by Fixing V_{GS}
Biasing by Fixing V_G and Connecting a Resistance in the Source
Biasing Using a Drain-to-Gate Feedback Resistor
Biasing Using a Constant-Current Source

Biasing in MOS Amplifier Circuits

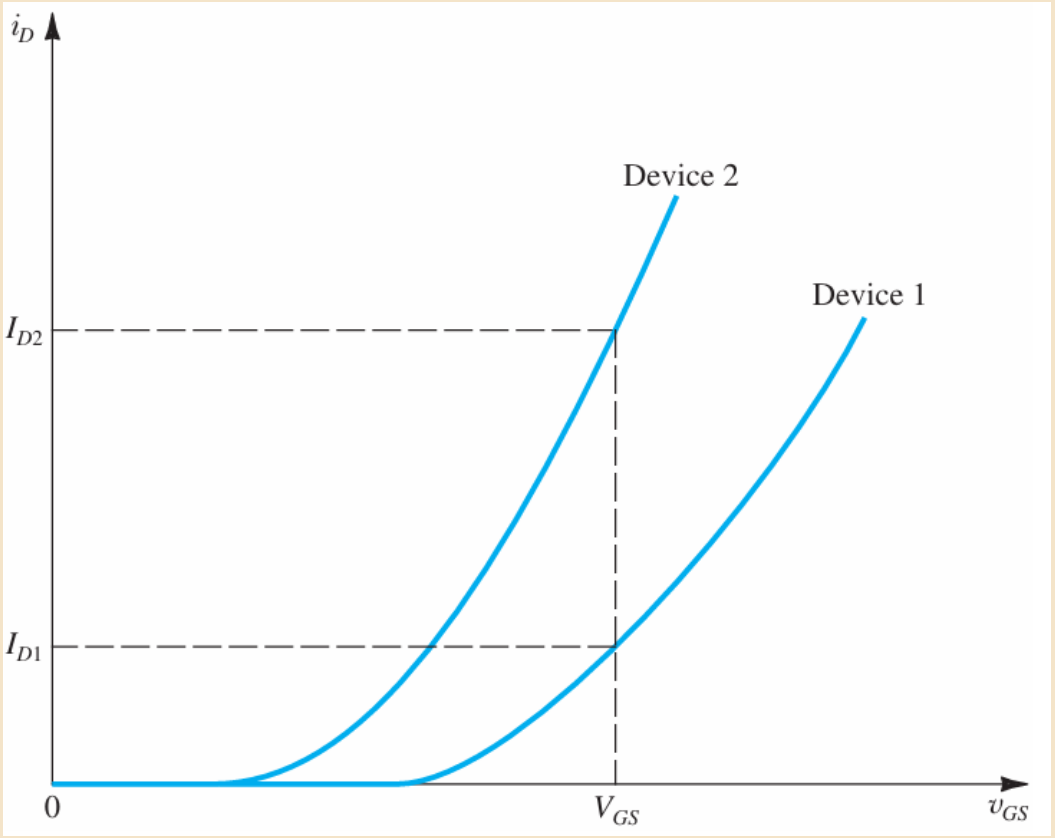
Biasing by Fixing V_{GS}:

The most straightforward approach to biasing a MOSFET is to fix its gate-to-source voltage V_{GS} to the value required to provide the desired $\boldsymbol{I_D}$ but this is not a good approach.

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_t)^2$$

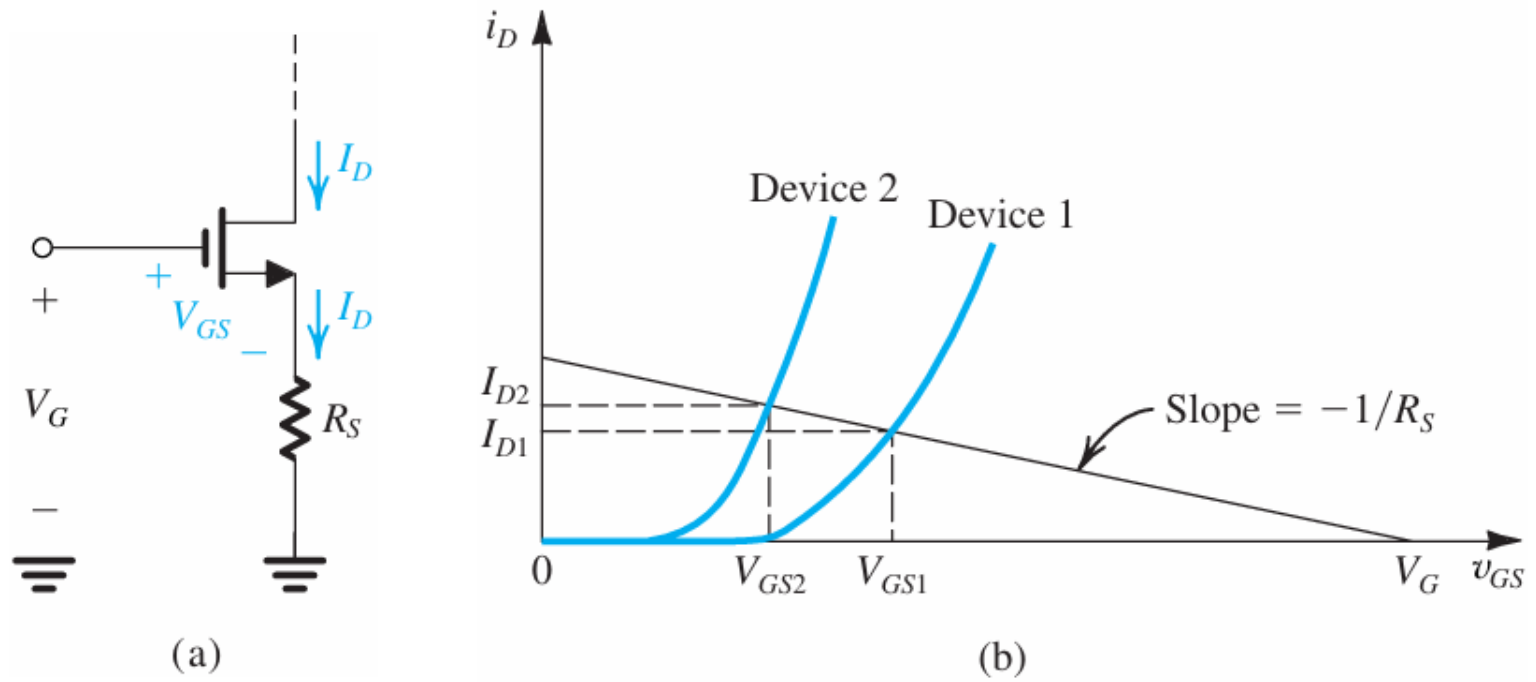
The parameters such as Threshold voltage V_t, the oxide-capacitance C_{ox} , the transistor aspect ratio W/L vary widely among devices of same size and type even they are from the same manufacturer.

Therefore Fixing V_{GS} does not ensure a constant drain current because MOSFET characteristics vary from device to device. Hence, this biasing method provides poor Q-point stability.



Biasing by Fixing V_G and Connecting a Resistance in the Source

Biasing by fixing V_G and connecting a resistor R_S in the source is a better method because it provides negative feedback and improves the stability of the operating point (Q-point).



$$V_G = V_{GS} + R_S I_D$$

The gate voltage V_G is fixed, and a resistor R_S is connected between the source and ground. If the drain current I_D increases, the source voltage $V_S = I_D R_S$ increases. Consequently, $V_{GS} = V_G - V_S$ decreases, reducing the drain current.

Fig. (b) provides a graphical illustration of the effectiveness of this biasing scheme. Here i_D–v_{GS} characteristics for two devices that represent the extremes of a batch of MOSFETs are shown. Superimposed on the device characteristics is a straight line that represents the constraint imposed by the bias circuit, represented by the equation inter section of this straight line with the i_D–v_{GS} characteristic curve provides the coordinates (I_D and V_{GS}) of the bias point. Observe that compared to the case of fixed V_{GS}, here the variability obtained in I_D is much smaller. Also, note that the variability decreases as V_G and R_S are made larger.

Thus the action of R_S works to keep I_D as constant as possible. This negative feedback action of R_S gives it the name **degeneration resistance.**

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